

MATH 3210-004 Selected Solutions to PSet Problems

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1 Introduction

In this document you may find solutions (proofs or examples) to selected problems from problem sets. As a general rule of thumb, it is more likely that this document will be updated with the solutions to problems that were graded on accuracy.

Throughout this class your solutions need not match the level of polish, precision and detail of the below solutions. What's most important is grasping the underlying concepts and steadily improving your problem-solving skills as well as your proof-writing and presentation skills.

Most solutions will have follow-up exercises. Just as in the problem sets, these follow-up exercises are stated in a neutral language, and determining whether or not the statement is correct is part of the exercise. A follow-up exercise here may later show up in an exam.

2 PSet 1, Problem 3

Let X and Y be sets. Let R be a relation from X to Y . Then

- (i) The relation $\mathcal{R} = \{ (A, B) \mid B = R(A) \}$ from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$ is the graph of a function.
- (ii) For any $A, B \subseteq X$, $R(A \cup B) = R(A) \cup R(B)$.
- (iii) For any $A, B \subseteq X$, $R(A \cap B) = R(A) \cap R(B)$.
- (iv) For any $A \subseteq X$, $R(A^c) = R(A)^c$.
- (v) For any $A, B \subseteq X$, $R(A \setminus B) = R(A) \setminus R(B)$.

We'll study each statement separately. The first statement is true:

Claim: The relation $\mathcal{R} = \{ (A, B) \mid B = R(A) \}$ from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$ is the graph of a function. ┐

Proof: We need to verify that the relation $\mathcal{R}$ from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$ defined by

$$\mathcal{R} = \{(A, B) \mid B = R(A)\} \subseteq \mathcal{P}(X) \times \mathcal{P}(Y)$$

is the graph of a function with domain $\mathcal{P}(X)$. For $A \in \mathcal{P}(X)$, by definition $R(A) \in \mathcal{P}(Y)$, so that indeed $\text{dom}(\mathcal{R}) = \mathcal{P}(X)$. Next let $(A_1, B_1), (A_2, B_2) \in \mathcal{R}$. Suppose $A_1 = A_2$. We need to show that $B_1 = B_2$. As $(A_1, B_1), (A_2, B_2) \in \mathcal{R}$, by the definition of $\mathcal{R}$, we have that

$$B_1 = R(A_1), \quad B_2 = R(A_2).$$

To show the equality of two sets, it suffices to show that either one is a subset of the other one. So let $y \in B_1 = R(A_1)$. Then for some $x \in A_1$, $(x, y) \in R$. But as $A_1 = A_2$, $x \in A_2$ also, and since $(x, y) \in R$,

we have that $y \in R(A_2) = B_2$. Thus $B_1 \subseteq B_2$. Interchanging the roles of B_1 and B_2 in this argument, we immediately obtain also $B_1 \supseteq B_2$, whence $B_1 = B_2$. In other words, $\mathcal{R}$ satisfies the abstract vertical line test, and consequently is the graph of a function from $\mathcal{P}(X)$ to $\mathcal{P}(Y)$.

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The second statement is also true:

Claim: For any $A, B \subseteq X$, $R(A \cup B) = R(A) \cup R(B)$.

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Proof: Let $A, B \subseteq X$. Then as $A \subseteq A \cup B$, $R(A) \subseteq R(A \cup B)$. Indeed, if $y \in R(A)$, there is an $x \in A$ such that $(x, y) \in R$. But $A \subseteq A \cup B$, so that $x \in A \cup B$, thus $y \in R(A \cup B)$. Similarly $R(B) \subseteq R(A \cup B)$. Then $R(A) \cup R(B) \subseteq R(A \cup B)$.

For the other direction, let $y \in R(A \cup B)$. Then there is an $x \in A \cup B$ such that $(x, y) \in R$. By definition $x \in A$ or $x \in B$, so that $y \in R(A)$ or $y \in R(B)$, that is, $R(A \cup B) \subseteq R(A) \cup R(B)$.

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The remaining three statements are false in general. To prove that a given statement is false, it suffices to provide one counterexample to the statement. For instance, the third statement is: "The image of the intersection of two subsets is equal to the intersection of the images". The **logical negation** of this is the statement "The image of the intersection of two subsets may be different than the intersection of the images". Using logical symbols, we may succinctly write:

$$\begin{aligned} & \neg[\forall A, B \in \mathcal{P}(X) : R(A \cap B) = R(A) \cap R(B)] \\ & \equiv \exists A, B \in \mathcal{P}(X) : R(A \cap B) \neq R(A) \cap R(B). \end{aligned}$$

We'll prove that the given statement is false, that is, the negation

of the given statement if true. As this latter is an existential statement, one example for it is sufficient; the example for the negation is a counterexample for the given statement.

Claim: The image of the intersection of two subsets under a relation is not necessarily equal to the intersection of the images.

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Proof: Consider $X = [0, 1] = Y$, $A = [0, 1/4]$, $B = [3/4, 1]$, $f : X \rightarrow Y$, $x \mapsto 1/2$, that is, f is the **constant function** that is constantly $1/2$. Then $A \cap B = \emptyset$, hence $f(A \cap B) = \emptyset$, while $f(A) = f(B) = \{1/2\}$, so that $f(A) \cap f(B) = \{1/2\}$.

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To obtain a correct version of the third statement, either we can weaken the conclusion, or strengthen the hypotheses¹.

Claim: The following version of the third statement with weakened conclusion is true:

$$\forall A, B \in \mathcal{P}(X) : R(A \cap B) \subseteq R(A) \cap R(B).$$

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Proof: Let $A, B \subseteq X$. As $A \cap B \subseteq A$, $R(A \cap B) \subseteq R(A)$. Similarly as $A \cap B \subseteq B$, $R(A \cap B) \subseteq R(B)$. Thus $R(A \cap B) \subseteq R(A) \cap R(B)$.

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Claim: The following version of the third statement with an additional hypothesis is true:

¹These are instances of logical **perturbations** of the original statement given; you do not need to submit such corrected versions, though trying to come up with such corrected versions will help you understand the concepts and prepare you for the exams better.

$$\begin{aligned} & [\forall (x_1, y_1), (x_2, y_2) \in R : y_1 = y_2 \implies x_1 = x_2] \\ \iff & [\forall A, B \in \mathcal{P}(X) : R(A \cap B) = R(A) \cap R(B)]. \end{aligned}$$

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Proof: ($\implies$) Suppose R satisfies

$$\forall (x_1, y_1), (x_2, y_2) \in R : y_1 = y_2 \implies x_1 = x_2;$$

one might call this the **injectivity condition** (aka **abstract horizontal line test**) for relations. Let $A, B \subseteq X$, and $y \in R(A) \cap R(B)$. Then there are $a \in A$ and $b \in B$ such that $(a, y), (b, y) \in R$. By the injectivity condition, this forces that $a = b$, and consequently that this element is in $A \cap B$. Thus $y \in R(A \cap B)$, so that $R(A) \cap R(B) \subseteq R(A \cap B)$. Since $R(A) \cap R(B) \supseteq R(A \cap B)$ for any relation R by the previous claim, we have $R(A) \cap R(B) = R(A \cap B)$.

($\impliedby$) Suppose R fails the injectivity condition, so that there are $x_1, x_2 \in X$ and $y \in Y$ such that $x_1 \neq x_2$ and $(x_1, y), (x_2, y) \in R$. Define $A = \{x_1\}$, $B = \{x_2\}$. Then $A \cap B = \emptyset$, so $R(A \cap B) = \emptyset$, but $y \in R(A) \cap R(B)$, that is, $R(A \cap B) \subsetneq R(A) \cap R(B)$.

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Claim: The image of the absolute complement of a subset under a relation need not be equal to the absolute complement of the image.

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Proof: We may repurpose our previous counterexample to establish this claim also. Put $X = Y = [0, 1]$, $A = [0, 1/4]$, $f : X \rightarrow Y$, $x \mapsto 1/2$. Then $A^c =]1/4, 1]$, $f(A^c) = \{1/2\}$, but $f(A)^c = \{1/2\}^c$, hence in fact we have that the two subsets are **disjoint**.

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Claim: The image of the relative complement of two subsets under a relation need not be equal to the relative complement of the images.

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Proof: Recalling that $A \setminus B = A \cap B^c$, the counterexample we gave earlier serves as a counterexample for this claim also. Indeed, put $X = Y = [0, 1]$, $A = [0, 1/4]$, $B = [3/4, 1]$, $f : X \rightarrow Y$, $x \mapsto 1/2$. Then $A \cap B^c = [0, 1/4]$, hence $f(A \setminus B) = \{1/2\}$; on the other hand $f(A) = f(B) = \{1/2\}$, so that $f(A) \setminus f(B) = \emptyset$.

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2.1 Follow-up Exercises

- (i) The second and the (correct versions) of the last third statements hold for arbitrary collections $\mathcal{A} \subseteq \mathcal{P}(X)$.
- (ii) The fourth and fifth statement has correct versions (one involving a "subset of" relation, and one involving the injectivity condition; similar to the correct versions of the third statement).
- (iii) If $f : X \rightarrow Y$ is a function, then the **preimage relation** f^{-1} satisfies the injectivity condition². Consequently, taking preimages **commutes** with unions, intersections and relative complements.

²Recall that f^{-1} really is an abbreviation for $\text{graph}(f)^{-1}$, and f^{-1} is not necessarily a function from Y to X .

3 PSet 1, Problem 6

Let X and Y be two sets. A function from X to Y is **surjunctive** if it is either surjective or non-injective. Then

- (i) X is finite iff any function from X to X is surjunctive.
- (ii) Y has exactly one element iff any function from X to Y is surjunctive.

The first claim is correct without any modifications, while the second statement is not quite correct as given. Still, a minor adjustment suffices to obtain a correct version. We start with the first claim.

For this claim, part of the challenge really is to make intuitive sense of what it means for a set to be finite³. From a set theoretical point of view, this calls for a **rigorous definition**, however we'll keep the notion of finiteness of sets at the naive level⁴. Thus for us, a set X is **finite** if it is either the emptyset or if it can be represented as

$$X = \{x_1, x_2, \dots, x_n\}$$

for some positive integer n . Similarly, a set X is **infinite** if for any positive integer n , there is an element $x_n \in X$ with the property that $x_i \neq x_j$ for $i \neq j$. With these definitions, for instance, it is true that saying that a set is not infinite is equivalent to saying that the set is finite, although proving this equivalence rigorously is surprisingly cumbersome! As we are really focused toward calculus, we'll not investigate this issue any further.

Claim: Let X be a set. Then X is finite iff for any $f \in F(X; X)$, f is

³In *this* sentence, we use the word "intuitive" intuitively and not in a **technical** sense. Throughout this course we'll almost never use "intuition" in a technical sense.

⁴Indeed, it is valid in the grand scheme of things to take the claim we set out to prove as the *definition* of what it means for a set to be finite.

surjunctive.

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Proof: Recall that "iff" stands for a necessary and sufficient condition. Thus we'll need to prove two implications.

($\Rightarrow$) First suppose $X = \{x_1, x_2, \dots, x_n\}$ is a finite set with exactly n elements and let $f : X \rightarrow X$ be an arbitrary function. We claim that f is surjunctive. That is, either f is surjective, or it is non-injective. To prove this, we may assume f fails to have one of these properties and show that it must have the other property. To that end, suppose f is injective (so that it fails to be non-injective). We need to show that f is surjective.

Suppose for a contradiction that there is a $j \in \{1, 2, \dots, n\}$ such that $x_j \notin \text{im}(f)$. But then $\text{im}(f)$ has at most $n - 1$ distinct elements. On the other hand, by the injectivity of f , for any $i, j \in \{1, 2, \dots, n\}$ with $i \neq j$, $f(x_i) \neq f(x_j)$, so that $\text{im}(f)$ has at least n distinct elements, a contradiction⁵.

($\Leftarrow$) For the **converse**, we'll argue contrapositively. Suppose X is infinite, so that we have for any positive integer n , an element $x_n \in X$ with the property that $x_i \neq x_j$ for $i \neq j$. We may then define a self-function of X as follows:

$$f : X \rightarrow X, x \mapsto \begin{cases} x_{n+1}, & \text{if } \exists n \in \mathbb{Z}_{\geq 1} : x = x_n \\ x, & \text{else} \end{cases}.$$

Thus f is the function that shifts the index of the specified indexed elements by one, and acts as identity on $X \setminus \{x_n | n \in \mathbb{Z}_{\geq 1}\}$. Then by construction, $x_1 \notin \text{im}(f)$, however f is injective, so that f fails to be surjunctive.

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⁵Alternatively, one can refer to the **pigeonhole principle** to derive a contradiction.

The second claim is not correct in general. Indeed, say $X = \{x_1, x_2, x_3\}$, $Y = \{y_1, y_2\}$. Then there are no injections from X to Y , thus any function $X \rightarrow Y$ is automatically surjunctive. The best we can do is to verify one side of the equivalence:

Claim: Let X and Y be two sets. If Y has exactly one element then any function $f : X \rightarrow Y$ is surjunctive.

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Proof: If $X = \emptyset$, then there are no functions $X \rightarrow Y$, so we may assume that $X \neq \emptyset$. Put $Y = \{y\}$ and let $f : X \rightarrow Y$ be function. Then f must be surjective, as there is at least one $x \in X$, and $f(x)$ is an element of Y , and consequently $f(x) = y$. Then by definition, f being surjective implies it being surjunctive.

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3.1 Follow-up Exercises

- (i) Let Y be a set. Then Y has exactly one element iff for any set X , any function in $F(X; Y)$ is surjunctive.

4 PSet 2, Problem 1

$$\forall n \in \mathbb{Z}_{\geq 1} : \sum_{k=1}^n k^3 = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2.$$

The statement given is correct without any modifications. We'll give both an induction proof and a direct proof. For brevity let us first set up some notation. Put

$$S : \mathbb{Z}_{\geq 1} \times \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}, (n, p) \mapsto \sum_{k=1}^n k^p.$$

Thus we have, for any n , $S(n, 0) = n$ and $S(n, 1) = \frac{n(n+1)}{2}$ (we proved this latter formula in class).

Claim:

$$\forall n \in \mathbb{Z}_{\geq 1} : S(n, 3) = \frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2.$$

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Proof (By Induction): For the base step, we need to verify that the statement holds for $n = 1$. Indeed; $S(1, 3) = 1$ and

$$\left[\frac{1}{4}n^4 + \frac{1}{2}n^3 + \frac{1}{4}n^2 \right] \Big|_{n=1} = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1.$$

For the induction step, let $n_* \in \mathbb{Z}_{\geq 1}$ be arbitrary and suppose the statement holds for $n = n_*$, that is,

$$S(n_*, 3) = \frac{1}{4}n_*^4 + \frac{1}{2}n_*^3 + \frac{1}{4}n_*^2. \quad (*)$$

We need to show that the statement holds for $n = n_* + 1$. We'll start with the polynomial function on the RHS and through some algebraic

identities rewrite it in terms of $S(n_*, 3)$; specifically we'll make use of the **binomial formula**:

$$\begin{aligned}
 & \frac{1}{4}(n_* + 1)^4 + \frac{1}{2}(n_* + 1)^3 + \frac{1}{4}(n_* + 1)^2 \\
 &= \frac{1}{4}(n_*^4 + 4n_*^3 + 6n_*^2 + 4n_* + 1) \\
 & \quad + \frac{1}{2}(n_*^3 + 3n_*^2 + 3n_* + 1) \\
 & \quad + \frac{1}{4}(n_*^2 + 2n_* + 1) \\
 &= \left(\frac{1}{4}n_*^4 + \frac{1}{2}n_*^3 + \frac{1}{4}n_*^2 \right) \\
 & \quad + \left(\frac{4}{4} \right) n_*^3 \\
 & \quad + \left(\frac{6}{4} + \frac{3}{2} \right) n_*^2 \\
 & \quad + \left(\frac{4}{4} + \frac{3}{2} + \frac{2}{4} \right) n_* \\
 & \quad + \left(\frac{1}{4} + \frac{1}{2} + \frac{1}{4} \right) \\
 &\stackrel{*}{=} S(n_*, 3) + (n_*^3 + 3n_*^2 + 3n_* + 1) \\
 &= S(n_*, 3) + (n_* + 1)^3 \\
 &= S(n_* + 1, 3),
 \end{aligned}$$

where in the equality with $*$ we used the induction hypothesis. Thus by the principle of mathematical induction, we are done.

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Proof (Direct Proof): For a direct proof, we'll relate the sum of p -th powers to the sums of lesser powers. Let $k, p \in \mathbb{Z}_{\geq 1}$ ⁶. Then

⁶When reading what follows, it might be helpful to take $p = 3$ to parse the manipulations.

$$(k+1)^{p+1} = \sum_{i=0}^{p+1} \binom{p+1}{i} k^i = k^{p+1} + (p+1)k^p + \sum_{i=0}^{p-1} \binom{p+1}{i} k^i$$

$$\implies k^p = \frac{(k+1)^{p+1} - k^{p+1}}{p+1} - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} k^i,$$

where for $\alpha, \beta \in \mathbb{Z}_{\geq 1}$ with $\alpha \geq \beta$,

$$\binom{\alpha}{\beta} = \frac{\alpha!}{\beta!(\alpha-\beta)!} = \frac{\alpha(\alpha-1)\cdots(\alpha-\beta+1)}{\beta(\beta-1)\cdots 2} \in \mathbb{Z}.$$

Hence for $n \in \mathbb{Z}_{\geq 1}$ we have

$$\begin{aligned} S(n, p) &= \sum_{k=1}^n k^p \\ &= \sum_{k=1}^n \left(\frac{(k+1)^{p+1} - k^{p+1}}{p+1} - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} k^i \right) \\ &= \frac{1}{p+1} \sum_{k=1}^n [(k+1)^{p+1} - k^{p+1}] - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} \left(\sum_{k=1}^n k^i \right) \\ &= \frac{1}{p+1} \sum_{k=1}^n [(k+1)^{p+1} - k^{p+1}] - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} S(n, i) \\ &= \frac{(n+1)^{p+1} - 1}{p+1} - \frac{1}{p+1} \sum_{i=0}^{p-1} \binom{p+1}{i} S(n, i). \end{aligned}$$

We have that $S(n, 0) = n$ by definition. Thus plugging in $p = 1$ in this latter formula, we obtain

$$S(n, 1) = \frac{(n+1)^2 - 1}{2} - \frac{1}{2}n = \frac{n^2 + n}{2},$$

(thus giving another direct proof for the formula for the sum of first powers). Next we plug in $p = 2$ to obtain:

$$\begin{aligned}
 S(n, 2) &= \frac{(n+1)^3 - 1}{3} - \frac{1}{3} \sum_{i=0}^1 \binom{3}{i} S(n, i) \\
 &= \frac{(n+1)^3 - 1}{3} - \frac{1}{3} (S(n, 0) + 3S(n, 1)) \\
 &= \frac{(n+1)^3 - 1}{3} - \frac{1}{3} (S(n, 0) + 3S(n, 1)) \\
 &= \frac{(n+1)^3 - 1}{3} - \frac{1}{3} S(n, 0) - S(n, 1) \\
 &= \frac{(n+1)^3 - 1}{3} - \frac{n}{3} - \frac{n^2 + n}{2} \\
 &= \frac{1}{3}n^3 + \frac{1}{2}n^2 + \frac{1}{6}n.
 \end{aligned}$$

Finally taking $p = 3$ and substituting the formulas we obtained so far, we end up with the formula for the cubes of integers; we omit the routine algebraic manipulations.

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Note that while the direct proof seems longer and more complicated than the proof by induction, it does not require an explicit formula and hence proves more than what is claimed, and further it signifies a method for a direct proof for any p .

4.1 Follow-up Exercises

- (i) For any $p \in \mathbb{Z}_{\geq 1}$, $S(\bullet, p)$ is a **polynomial** of degree $p + 1$ with rational coefficients.
- (ii) Let $\alpha(p), \mu(p)$ be the minimal numbers of rational number additions and rational number multiplications one would need to execute to compute all the coefficients of $S(\bullet, p)$, respectively, for p an arbitrary positive integer.
 - (a) Compute $\alpha(p)$ and $\mu(p)$ for $p \in \{0, 1, 2\}$.
 - (b) Find two functions $\hat{\alpha}$ and $\hat{\mu}$ that can be written as the composition of finitely many **elementary functions** with the property that for some $C \in \mathbb{R}_{>0}$ and some $P \in \mathbb{Z}_{\geq 1}$:

$$\forall p \in \mathbb{Z}_{\geq P} : \quad \alpha(p) \leq C\hat{\alpha}(p), \quad \mu(p) \leq C\hat{\mu}(p).$$

- (c) Find the **best** functions $\hat{\alpha}$ and $\hat{\mu}$ that have the properties in the previous part.

5 PSet 2, Problem 2

Let $A \subseteq \mathbb{R}$ be nonempty and bounded. Then

- (i) The set of all upper bounds of A is $[\sup(A), \infty[$.
- (ii) The set of all lower bounds of A is $] - \infty, \inf(A)]$.
- (iii) The smallest closed interval containing A is

$$[\inf(A), \sup(A)].$$

- (iv) The largest open interval contained in A is

$$] \inf(A), \sup(A)[.$$

Let $A \subseteq \mathbb{R}$. First let us set some notation for convenience. Define by $\mathcal{U}(A)$, $\mathcal{L}(A)$ the sets of all upper and lower bounds of A , respectively:

$$\mathcal{U}(A) = \{U \in \mathbb{R} \mid \forall a \in A : a \leq U\},$$

$$\mathcal{L}(A) = \{L \in \mathbb{R} \mid \forall a \in A : L \leq a\}.$$

We first establish that the first three statements are correct without modification:

Claim: Let $A \subseteq \mathbb{R}$ be nonempty and bounded, then

- (i) $\mathcal{U}(A) = [\sup(A), \infty[\subseteq \mathbb{R}$,
- (ii) $\mathcal{L}(A) =] - \infty, \inf(A)] \subseteq \mathbb{R}$,
- (iii) The smallest closed interval containing A is $[\inf(A), \sup(A)] \subseteq \mathbb{R}$.

- (iv) The largest open interval contained in A is $] \inf(A), \sup(A)[$ iff A is an **interval**.

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Proof: We are given that A is bounded and nonempty, hence we have that neither $\mathcal{U}(A)$ nor $\mathcal{L}(A)$ are empty. In particular, by the completeness axiom we have that the minimum of $\mathcal{U}(A)$ as well as the maximum of $\mathcal{L}(A)$ exist; $\sup(A)$ and $\inf(A)$ are *defined* to be these real numbers.

(i) and (ii) Note that if U is an upper bound, then any number larger than U too is an upper bound, and similarly one can replace a lower bound by an even lower number. Thus we have:

$$\mathcal{U}(A) = \bigcup_{U \in \mathcal{U}(A)} [U, \infty[, \quad \mathcal{L}(A) = \bigcup_{L \in \mathcal{L}(A)}] - \infty, L]. \quad (*)$$

In addition, if U is an upper bound of A , then $\sup(A) \leq U$, that is, $U \in [\sup(A), \infty[$, so that $\mathcal{U}(A) = [\sup(A), \infty[$. Similarly if L is a lower bound of A , L must be in $] - \infty, \inf(A)]$, so that $\mathcal{L}(A) \subseteq] - \infty, \inf(A)]$. This concludes the proof of the first two parts.

(iii) We may argue as follows for the third part: By definition $\inf(A)$ is a lower bound for any element in A and similarly $\sup(A)$ is an upper bound for any element in A , thus $A \subseteq [\inf(A), \sup(A)]$. If $[\lambda, \rho] \subseteq \mathbb{R}$ were a closed and bounded interval containing A , then λ would be a lower bound and ρ would be an upper bound for A ; hence by definition $\lambda \leq \inf(A)$ and $\sup(A) \leq \rho$. That is $[\inf(A), \sup(A)] \subseteq [\lambda, \rho]$. In other words, $[\inf(A), \sup(A)]$ indeed is the **smallest** closed interval containing A .

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Finally we study the last statement. In general the last statement

fails, indeed, for instance for $A = \{0, 1\}$, a set with exactly two elements, $1/2$ is in $]0, 1[=]\inf(A), \sup(A)[$ ⁷.

Still, if we assume that A is an interval to begin with, the fourth statement becomes true. Recall that the definition we are using for A to be an interval is that it satisfies the following property:

$$\forall x, y \in A, \forall t \in [0, 1] : (1 - t)x + ty \in A.$$

Claim: Let $A \subseteq \mathbb{R}$ be nonempty and bounded, then A is an interval iff the largest open interval contained in A is $] \inf(A), \sup(A)[$.

┘

Proof: Let A be nonempty and bounded. We have two cases: either $\inf(A) = \sup(A)$ xor $\inf(A) < \sup(A)$. In the first case, we have that A is a singleton, and consequently it does not contain any nonempty open interval, while it trivially satisfies the condition for being an interval, so that there is nothing to do. Suppose therefore that $\inf(A) < \sup(A)$.

($\Rightarrow$) Suppose A is a nonempty and bounded interval and let $\mu \in] \inf(A), \sup(A)[$. Then μ fails to be a lower bound as well as an upper bound of A , by the definitions of infimum and supremum, respectively. Thus there are $x, y \in A$ such that

$$\inf(A) \leq x < \mu < y \leq \sup(A).$$

Since A is an interval and $\mu = (1 - t_*)x + t_*y$ for $t_* = \frac{\mu - x}{y - x} \in]0, 1[$, $\mu \in A$. Therefore $] \inf(A), \sup(A)[\subseteq A$.

Let $] \lambda, \rho[$ be an arbitrary open interval contained in A . **WLOG** we may assume that this interval is nonempty and take $\mu \in] \lambda, \rho[$. Then $\mu \in A$ and hence $\mu \in [\inf(A), \sup(A)]$. It must be the case that $\mu > \inf(A)$, for otherwise, $\frac{\lambda + \inf(A)}{2}$, a number strictly less than $\inf(A)$, would

⁷For the purposes of problem set grading, concluding here would have been sufficient.

be in A , a contradiction. Similarly it must be the case that $\mu \leq \sup(A)$, whence $]\lambda, \rho[\subseteq]\inf(A), \sup(A)[$, so that indeed $]\inf(A), \sup(A)[$ is the largest open interval contained in A .

($\Leftarrow$) We'll argue by contradiction. Suppose A is bounded and nonempty, that $]\inf(A), \sup(A)[$, but that it fails to be an interval. Then there are $x, y \in A$ and $t_* \in]0, 1[$ such that $x \leq y$ and $(1 - t_*)x + t_*y \notin A$. But then

$$(1 - t_*)x + t_*y \in]x, y[\subseteq]\inf(A), \sup(A)[\subseteq A,$$

a contradiction. ┘

We make a final remark to contextualize this problem: We had discussed the **"clamp" heuristic for supremum and infimum in class**, and the problem really is a formalization of this heuristic: increasing L in $]-\infty, L]$ corresponds to tightening the lower clamp, decreasing U in $[U, \infty[$ corresponds to tightening the upper clamp, and the numbers L and U that correspond to the tightest possible configuration are the infimum and supremum, respectively. The heuristic works for clamping from without. While there are physical devices that **clamp from within**, they are different in principle than the abstract notions on infimum and supremum, whence the fourth part is incorrect unless we have additional information regarding the nature of the set we are trying to clamp down on.

6 PSet 2, Problem 3

Let $A, B \subseteq \mathbb{R}$. Then

$$(i) \sup(A \cup B) = \max\{\sup(A), \sup(B)\}$$

$$(ii) \inf(A \cup B) = \min\{\inf(A), \inf(B)\}.$$

We separate cases as usual: either at least one of A and B is empty, xor neither A nor B is empty.

The $A = \emptyset$ or $B = \emptyset$ Case WLOG, we may assume $A \neq \emptyset = B$. Then $A \cup B = A$ and

$$\sup(A \cup B) = \sup(A) > -\infty = \sup(\emptyset) = \sup(B),$$

$$\inf(A \cup B) = \inf(A) < \infty = \inf(\emptyset) = \inf(B).$$

Since the maximum of an extended real number R and $-\infty$ is R by definition, the first part follows. Similarly since the minimum of an extended real number R and ∞ is R by definition the second part is also true.

The $A \neq \emptyset \neq B$ Case For the first part, WLOG we may assume that $\sup(A) \geq \sup(B)$. Thus it suffices to show that $\sup(A) = \sup(A \cup B)$. First note that as $A \subseteq A \cup B$, by the **monotonicity** of supremum we have

$$-\infty \leq \sup(B) \leq \sup(A) \leq \sup(A \cup B) \leq \infty.$$

Either $A \cup B$ is unbounded from above, xor it is bounded from above. If $A \cup B$ is unbounded from above, then at least one of A or

B would have to be unbounded from above. If A is unbounded from above, then $\sup(A) = \infty = \sup(A \cup B)$ concludes the proof. If B is unbounded from above, then $\infty = \sup(B) \leq \sup(A) \leq \sup(A \cup B) \leq \infty$, hence again $\sup(A) = \sup(A \cup B)$ (and consequently A is unbounded from above also).

Now suppose $A \cup B$ is bounded from above; then the same is true for both A and B, and we have

$$-\infty \leq \sup(B) \leq \sup(A) \leq \sup(A \cup B) \leq \infty.$$

Say $\sup(A) < \sup(A \cup B)$. Then $\sup(A)$ is not an upper bound for $A \cup B$, so that there is an $x \in A \cup B$ such that

$$\sup(A) < x.$$

Since x exceeds the least upper bound for A, $x \notin A$, thus it must be the case that $x \in B$. But then

$$x \leq \sup(B) \leq \sup(A) < x;$$

a contradiction.

One can argue analogously for the statement involving infima. Alternatively one can argue as follows⁸:

$$\begin{aligned} \inf(A \cup B) &= -\sup(-(A \cup B)) \\ &= -\sup((-A) \cup (-B)) \\ &= -\max\{\sup(-A), \sup(-B)\} \\ &= -\max\{-\inf(A), -\inf(B)\} \\ &= \min\{\inf(A), \inf(B)\}, \end{aligned}$$

⁸You should check these steps if they are not clear!

where $-C = \{-c \in \mathbb{R} \mid c \in C\}$ for $C \subseteq \mathbb{R}$ a subset.

We have thus proved that indeed the statements in the problem are correct without any modifications:

Claim: $\forall A, B \subseteq \mathbb{R}$:

(i) $\sup(A \cup B) = \max\{\sup(A), \sup(B)\}$

(ii) $\inf(A \cup B) = \min\{\inf(A), \inf(B)\}$.

┘

6.1 Follow-up Exercises

(i) $\forall \mathcal{A} \subseteq \mathcal{P}(\mathbb{R})$:

(a) $\sup \left(\bigcup_{A \in \mathcal{A}} A \right) = \sup_{A \in \mathcal{A}} \sup(A),$

(b) $\inf \left(\bigcup_{A \in \mathcal{A}} A \right) = \inf_{A \in \mathcal{A}} \inf(A).$

(ii) $\forall \mathcal{A} \subseteq \mathcal{P}(\mathbb{R})$:

(a) $\sup \left(\bigcap_{A \in \mathcal{A}} A \right) = \inf_{A \in \mathcal{A}} \sup(A),$

(b) $\inf \left(\bigcap_{A \in \mathcal{A}} A \right) = \sup_{A \in \mathcal{A}} \inf(A).$

7 PSet 3, Problem 2

Let $x_\bullet, y_\bullet$ be two convergent sequences of real numbers with limits $x_\infty, y_\infty \in \mathbb{R}$ respectively:

$$\lim_{n \rightarrow \infty} x_n = x_\infty, \quad \lim_{n \rightarrow \infty} y_n = y_\infty.$$

Then

- (i) $\lim_{n \rightarrow \infty} (x_n + y_n) = x_\infty + y_\infty$.
- (ii) For any real number α , $\lim_{n \rightarrow \infty} (\alpha x_n) = \alpha x_\infty$.
- (iii) $\lim_{n \rightarrow \infty} (x_n y_n) = x_\infty y_\infty$.
- (iv) $\lim_{n \rightarrow \infty} \max\{x_n, y_n\} = \max\{x_\infty, y_\infty\}$.

All these limit laws are indeed correct without any modifications. The last statement involving maxima is slightly trickier than the rest, so we'll study it separately.

Claim: Let $x_\bullet \rightarrow x_\infty, y_\bullet \rightarrow y_\infty$ be two convergent sequences of real numbers. Then

- (i) $\lim_{n \rightarrow \infty} (x_n + y_n) = x_\infty + y_\infty$.
- (ii) For any real number α , $\lim_{n \rightarrow \infty} (\alpha x_n) = \alpha x_\infty$.
- (iii) $\lim_{n \rightarrow \infty} (x_n y_n) = x_\infty y_\infty$.

┘

For statements like these, we'll need to take an arbitrary $\varepsilon \in \mathbb{R}_{>0}$ and find some finite time $N \in \mathbb{Z}_{\geq 0}$ such that after this time, the distance between the position at time n (n being a time later than N) and the claimed target at most ε . Perhaps surprisingly, instead of

starting with the ε , often it's more practical to start from the *end of the proof*. In other words, first with start with the distance we are trying to control, and then using some inequalities we establish that the distance we are trying to control can be controlled via the distances we are given control of. With all this said, a proof might go like this:

Proof: To be more concrete, let us start indexing the times of the sequences involved from 0⁹.

(i) For the first part, we have, for any n ,

$$|(x_n + y_n) - (x_\infty + y_\infty)| \leq |x_n - x_\infty| + |y_n - y_\infty| \quad (*)$$

by the triangular inequality. We know that each of the two summands on the RHS can be made arbitrarily small, and so this will suffice.

Let $\varepsilon \in \mathbb{R}_{>0}$. Then there are two times $N_1, N_2 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$n \geq N_1 \implies |x_n - x_\infty| < \varepsilon/2,$$

$$n \geq N_2 \implies |y_n - y_\infty| < \varepsilon/2.$$

But then, putting $N = \max\{N_1, N_2\}$, we have that for any $n \in \mathbb{Z}_{\geq 0}$, if $n \geq N$, then both of these bounds hold simultaneously. Thus for $n \geq N$, by (*), we have

$$|(x_n + y_n) - (x_\infty + y_\infty)| < \varepsilon.$$

(ii) We have for any $n \in \mathbb{Z}_{\geq 0}$,

⁹Ultimately at what time one starts indexing does not matter for the purposes of this proof, as the claim is all about the *future asymptotics*, insofar as the indices of different sequences are compatible.

$$|(\alpha x_n) - (\alpha x_\infty)| = |\alpha| |x_n - x_\infty| \leq (|\alpha| + 1) |x_n - x_\infty|. \quad (\dagger)$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then $\varepsilon/(|\alpha| + 1) \in \mathbb{R}_{>0}$ also, and hence there is a time $N \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$n \geq N \implies |x_n - x_\infty| < \varepsilon/(|\alpha| + 1).$$

Using this latter bound together with $(\dagger)$, we obtain that for any $n \geq N$,

$$|(\alpha x_n) - (\alpha x_\infty)| < \varepsilon.$$

(iii) This one is slightly more tricky. Whenever one tries to bound distances between "terms with factors", a good method is to "inject" some choice cross terms¹⁰. Thus we have, for any n ,

$$\begin{aligned} & |(x_n y_n) - (x_\infty y_\infty)| \\ &= |x_n y_n - x_n y_\infty + x_n y_\infty - x_\infty y_\infty| \\ &\leq |x_n y_n - x_n y_\infty| + |x_n y_\infty - x_\infty y_\infty| \\ &= |x_n| |y_n - y_\infty| + |x_n - x_\infty| |y_\infty| \\ &= |x_n - x_\infty + x_\infty| |y_n - y_\infty| + |x_n - x_\infty| |y_\infty| \\ &\leq (|x_n - x_\infty| + |x_\infty|) |y_n - y_\infty| + |x_n - x_\infty| |y_\infty|. \quad (\$) \end{aligned}$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then $\min\{1, \varepsilon/2(|y_\infty| + 1)\} \in \mathbb{R}_{>0}$ also¹¹, and since $x_\bullet \rightarrow x_\infty$, there is an $N_1 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

¹⁰Here by "inject" I mean adding and subtracting the same term and then using the triangular inequality.

¹¹As you are reading this proof, at this point you *should not* worry about why this number is relevant, and keep reading. For now, this is merely a possible error margin that depends on ε already chosen. See also the next footnote at the end of the proof.

$$n \geq N_1 \implies |x_n - x_\infty| < \min\{1, \varepsilon/2(|y_\infty| + 1)\}.$$

Further, $\varepsilon/2(|x_\infty| + 1) \in \mathbb{R}_{>0}$ too, and since $y_\bullet \rightarrow y_\infty$, there is an $N_2 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$n \geq N_2 \implies |y_n - y_\infty| < \varepsilon/2(|x_\infty| + 1).$$

Combining these two inequalities with (\$), we have that, putting $N = \max\{N_1, N_2\}$, for any $n \in \mathbb{Z}_{\geq 0}$,

$$n \geq N \implies |(x_n y_n) - (x_\infty y_\infty)| < \varepsilon.^{12}$$

┘

Claim ((iv)): Let $x_\bullet \rightarrow x_\infty, y_\bullet \rightarrow y_\infty$ be two convergent sequences of real numbers. Then

$$\lim_{n \rightarrow \infty} \max\{x_n, y_n\} = \max\{x_\infty, y_\infty\}.$$

┘

We'll provide two separate proofs of this claim; one is more straight-

¹²How did I come up with the error margins for $x_\bullet$ and $y_\bullet$, as functions of ε , so that when the dust settled I obtained the nice ε bound for what I ultimately wanted to control? You'll see such bizarre expressions in the middle of proofs in analysis quite often; and the answer almost always is this: the author of the proof again worked *backwards*, and did some calculations *not worthy of explanation* (supposedly). In such cases it's best to not dwell too much on the specific choices of numbers and to keep reading the proof (as mentioned in the previous footnote). For instance in this case I was really looking at the RHS in (\$). I want this term to be less than ε . As this is the sum of two terms, I decided to simply bound both summands by $\varepsilon/2$ each. The second summand has a factor that may or may not be 0, so I scaled $\varepsilon/2$ by dividing it by $|y_\infty| + 1 \in \mathbb{R}_{>0}$. But then I realized I'm going to need to bound the first term by $\varepsilon/2$ also, and this scaled version of $\varepsilon/2$ is too cumbersome to manipulate algebraically, hence I simply took the minimum of it with 1, in case the first supposed bound is relatively large etc.. There are many other bounds that would have worked, and ultimately none of this is really necessary. Instead you can simply bound the RHS by some function of ε that decays to zero as ε does. For instance in this case it's much more straightforward to bound the RHS by a quadratic polynomial in ε that has a zero at $\varepsilon = 0$ (such a polynomial decays like ε^2 , which for ε small enough, is even smaller than ε). There are *contexts* where one *needs* to be careful about sharp bounds (for instance if the problem was asking statements regarding the *rates of convergences*), but often there is a fair amount of flexibility with epsilon management.

forward but requires that one first establishes an inequality regarding maxima, while the other one is less straightforward but does not use any technology. For the first proof we first establish a lemma:

Lemma: For any $\alpha, \bar{\alpha}, \beta, \bar{\beta} \in \mathbb{R}$,

$$|\max\{\alpha, \bar{\alpha}\} - \max\{\beta, \bar{\beta}\}| \leq \max\{|\alpha - \beta|, |\bar{\alpha} - \bar{\beta}|\}.$$

┘

Proof: Wlog, we may assume $\alpha \leq \bar{\alpha}$ and $\beta \leq \bar{\beta}$. Then

$$|\max\{\alpha, \bar{\alpha}\} - \max\{\beta, \bar{\beta}\}| = |\bar{\alpha} - \bar{\beta}| \leq \max\{|\alpha - \beta|, |\bar{\alpha} - \bar{\beta}|\}.$$

┘

Proof (of Claim (iv), using Lemma): We have that, by the lemma, for any $n \in \mathbb{Z}_{\geq 0}$:

$$|\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| \leq \max\{|x_n - x_\infty|, |y_n - y_\infty|\}. \quad (\spadesuit)$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then there are $N_1, N_2 \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$:

$$n \geq N_1 \implies |x_n - x_\infty| < \varepsilon,$$

$$n \geq N_2 \implies |y_n - y_\infty| < \varepsilon.$$

Consequently, putting $N = \max\{N_1, N_2\}$, by $(\spadesuit)$, we have that for any $n \in \mathbb{Z}_{\geq 0}$,

$$n \geq N \implies |\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| < \varepsilon.$$

┘

Proof (of Claim (iv), without using Lemma): Wlog, we may assume that $x_\infty \leq y_\infty$. Then either $x_\infty \leq y_\infty$ xor $x_\infty = y_\infty$.

The $x_\infty \leq y_\infty$ Case In this case we have that $(y_\infty - x_\infty)/2 \in \mathbb{R}_{>0}$. Thus, using this positive number as the error margin, we have that there is a time $N \in \mathbb{Z}_{\geq 0}$ such that¹³ for any $n \in \mathbb{Z}_{\geq 0}$,

$$\begin{aligned} n \geq N \\ \implies |x_n - x_\infty| &< (y_\infty - x_\infty)/2 \\ \text{and } |y_n - y_\infty| &< (y_\infty - x_\infty)/2. \end{aligned}$$

This implies¹⁴ that

$$n \geq N \implies x_n \leq y_n.$$

Consequently, if $n \geq N$, we have

$$|\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| = |y_n - y_\infty|. \quad (\clubsuit)$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then there is an $N' \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq 0}$,

$$n \geq N' \implies |y_n - y_\infty| < \varepsilon.$$

Putting $N'' = \max\{N, N'\}$, using this last inequality together with $(\clubsuit)$, we have that for any $n \in \mathbb{Z}_{\geq 0}$,

¹³Here we are fast-forwarding the procedure of first finding two separate times for the two sequences to be sufficiently near their respective limits and then taking the maximum of these two times so as to guarantee simultaneity.

¹⁴Draw a picture, or rearrange the above inequalities!

$$n \geq N'' \implies |\max\{x_n, y_n\} - \max\{x_\infty, y_\infty\}| < \varepsilon.$$

The $x_\infty = y_\infty$ Case For clarity denote by L the common limit. Then both $x_\bullet$ and $y_\bullet$ converges to L . Let $\varepsilon \in \mathbb{R}_{>0}$. Then there is an $N \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq N}$,

$$|x_n - L| < \varepsilon,$$

$$|y_n - L| < \varepsilon.$$

Rearranging these inequalities, we obtain that for any $n \in \mathbb{Z}_{\geq N}$,

$$L - \varepsilon < x_n < L + \varepsilon,$$

$$L - \varepsilon < y_n < L + \varepsilon.$$

Note that the upperbound the maximum of two numbers, one needs to upperbound *both numbers involved*. Hence the RHS's of the inequalities above give us

$$\forall n \in \mathbb{Z}_{\geq N} : \max\{x_n, y_n\} < L + \varepsilon.$$

To lowerbound the maximum of two numbers, one needs to lowerbound *at least one of the numbers involved*. The LHS's of the inequalities above in fact are *more than enough* to conclude

$$\forall n \in \mathbb{Z}_{\geq N} : L - \varepsilon < \max\{x_n, y_n\}.$$

Combining these two last inequalities, we have:

$$\forall n \in \mathbb{Z}_{\geq N} : |\max\{x_n, y_n\} - L| < \varepsilon.$$

」

7.1 Follow-up Exercises

- (i) Here is one way to get more quantitative versions of the above results: Let $x_\bullet : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}$ be a convergent sequence. The limit of $x_\bullet$ is a uniquely determined real number, so we don't need to explicitly specify it. Then for any $\varepsilon \in \mathbb{R}_{>0}$, the following set of non-negative integers is nonempty:

$$Z_\varepsilon = \left\{ N \in \mathbb{Z}_{\geq 0} \mid \forall n \in \mathbb{Z}_{\geq N} : \left| x_n - \lim_{n \rightarrow \infty} x_n \right| < \varepsilon \right\}.$$

Then by the Well-Ordering Principle, this set has a unique minimum; let's denote this minimum by $\mathcal{N}_\varepsilon(x_\bullet)$. Thus $\mathcal{N}_\varepsilon(x_\bullet)$ is *exactly* how long one would need to wait for $x_\bullet$ to be within an ε of its limit. We can extend this definition to arbitrary sequences by setting $\mathcal{N}_\varepsilon(x_\bullet) = \infty$ if $Z_\varepsilon = \emptyset$, so that ultimately we have a function

$$\mathcal{N} : \mathbb{R}_{>0} \times F(\mathbb{Z}_{\geq 0}; \mathbb{R}) \rightarrow \mathbb{Z}_{\geq 0} \cup \{\infty\}, (\varepsilon, x_\bullet) \mapsto \mathcal{N}_\varepsilon(x_\bullet).$$

Then the above statements can be rephrased in terms of $\mathcal{N}$. For instance, the proof of the statement regarding the limit of the sums of two sequences can be rephrased as:

$$\forall \varepsilon \in \mathbb{R}_{>0} : \mathcal{N}_\varepsilon(x_\bullet + y_\bullet) \leq \max\{\mathcal{N}_{\varepsilon/2}(x_\bullet), \mathcal{N}_{\varepsilon/2}(y_\bullet)\}.$$

How about an *exact* formula for the time on LHS, for this part or the other parts? For instance a bound that seems a priori better is this:

$$\forall \varepsilon \in \mathbb{R}_{>0} : \mathcal{N}_\varepsilon(x_\bullet + y_\bullet) \leq \inf_{\eta \in]0,1[} \max\{\mathcal{N}_{(1-\eta)\varepsilon}(x_\bullet), \mathcal{N}_{\eta\varepsilon}(y_\bullet)\}.$$

- (ii) One can **generalize** the lemma above as follows: Let X be a set, $\alpha, \beta \in F(X; \mathbb{R})$. Then

$$|\sup \alpha - \sup \beta| \leq \sup |\alpha - \beta|.$$

- (iii) State and prove versions of the last statement with minima.

8 PSet 3, Problem 3

Let $x_\bullet$ be a sequence of real numbers that converges to zero.
Then for any bounded sequence $b_\bullet$ of real numbers,

$$\lim_{n \rightarrow \infty} x_n b_n = 0.$$

This claim is indeed true without any modifications:

Claim: $\forall x_\bullet, b_\bullet : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R} : \text{if } x_\bullet \rightarrow 0 \text{ and } \sup_m b_m < \infty, \text{ then } x_\bullet b_\bullet \rightarrow 0.$ ┘

Proof: Instead of starting with an $\epsilon > 0$, it's often useful to start from the end, in other words with the expression we are trying to bound. For any $n \in \mathbb{Z}_{\geq 0}$, one has

$$|x_n b_n| = |x_n| |b_n| \leq |x_n| \underbrace{\sup_{m \in \mathbb{Z}_{\geq 0}} |b_m|}_{< \infty} \quad (*)$$

Put $M = \sup_{m \in \mathbb{Z}_{\geq 0}} |b_m| + 1 \in \mathbb{R}_{\geq 0}$. We know that this is a finite nonnegative number, as the sequence $b_\bullet$ (and consequently the sequence $|b_\bullet|$) is bounded. We added 1 to the supremum of $|b_\bullet|$ to ensure that M is a positive number, so that dividing by M is well-defined¹⁵. Thus to bound $|x_n b_n|$ from above by ϵ , we can bound $|x_n|$ by ϵ/M . Indeed, let $\epsilon \in \mathbb{R}_{>0}$. Then as $x_\bullet \rightarrow 0$, there is an $N \in \mathbb{Z}_{\geq 0}$ such that for any $n \in \mathbb{Z}_{\geq N}$, we have

$$|x_n| = |x_n - 0| < \frac{\epsilon}{M}.$$

¹⁵Alternatively, we could have done a case analysis: either $b_\bullet$ is the sequence that is constantly zero, in which case there is nothing to do, or else $b_\bullet$ is not constantly zero, so that the supremum of the absolute value sequence $|b_\bullet|$ is strictly positive.

Here we used the number $\frac{\varepsilon}{M}$ as the "epsilon" in the definition of x_n converging to 0. For any $n \in \mathbb{Z}_{\geq N}$, $(*)$ also holds; hence combining the two we have that for any $n \in \mathbb{Z}_{\geq N}$:

$$|x_n b_n - 0| = |x_n b_n| < \frac{\varepsilon}{M}(M - 1) < \varepsilon.$$

┘

8.1 Follow-up Exercises

- (i) Let x_n be a sequence of real numbers. Then the following are equivalent:
- (a) x_n converges to 0.
 - (b) For any bounded sequence b_n , $x_n b_n$ converges to 0.
 - (c) For some bounded sequence b_n , $x_n b_n$ converges to 0.

9 PSet 4, Problem 1

Let I be a (nonempty) interval, $f \in F(I; \mathbb{R})$. Then the following are equivalent:

A: f is continuous.

B: For any $k \in \mathbb{Z}_{\geq 1}$ and any $x_1, \dots, x_k \in I$, there is an $x_{\dagger} \in I$ such that $f(x_{\dagger}) = \frac{f(x_1) + \dots + f(x_k)}{k}$.

Continuity implies the "averageability" property, but not vice versa in general. For I an interval and $f \in F(I; \mathbb{R})$, let's say that f has the **average value property** if for any finite nonempty subset $S \subseteq \text{im}(f)$, there is an $x_S \in I$ such that

$$f(x_S) = \frac{1}{\#(S)} \sum_{y \in S} y.$$

Similarly let's say that f has the **intermediate value property** (aka **Darboux property**) if for any subinterval $J \subseteq I$, $f(J)$ is an interval.

Thus the statement **B** above is equivalent to the average value property, and the Intermediate Value Theorem can be concisely stated like so: "The image of any interval under a real valued continuous function is an interval."

(Note that neither of these properties require any assumption on the domain X .)

Claim: Let I be a (nonempty) interval, $f \in F(I; \mathbb{R})$. Then:

- (i) If f has the intermediate value property, then it has the average value property.
- (ii) If f is continuous, then it has the average value property.

┘

Proof: The second item follows immediately from the first by way of the Intermediate Value Theorem. Thus it suffices to prove the first item. Suppose f has the intermediate value property, let $S \subseteq \text{im}(f)$ be a finite and nonempty set. Then for any $y \in S$,

$$\min(S) \leq y \leq \max(S).$$

Note that as S is a finite set, its maximum and minimum are well defined, and consequently are in the image of f . Adding all elements in S , and rearranging, we have

$$\frac{1}{\#(S)} \sum_{y \in S} y \in [\min(S), \max(S)] \subseteq \text{im}(f),$$

hence by the hypothesis that f satisfies the intermediate value property, there is at least one $x_S \in I$ such that

$$f(x_S) = \frac{1}{\#(S)} \sum_{y \in S} y,$$

as was to be shown. ┘

Here is another proof of the same claim using induction.

Proof (Using Induction): We'll use induction on $\#(S)$. If $\#(S) = 1$, then by definition there is nothing to do. Suppose k_* is a positive integer such that for any subset $S \subseteq \text{im}(f)$ with $\#(S) = k_*$, the statement is true. Let $T \subseteq \text{im}(f)$ be a subset with $\#(T) = k_* + 1$. Let $y_* \in T$ and consider the set $S = T \setminus \{y_*\}$. Then by induction hypothesis, there is an $x_* \in I$ such that

$$f(x_*) = \frac{1}{k_*} \sum_{y \in S} y.$$

Then

$$\begin{aligned}
 \frac{1}{k_* + 1} \sum_{y \in T} y &= \frac{1}{k_* + 1} \left(\sum_{y \in S} y + y_* \right) \\
 &= \frac{1}{k_* + 1} \left(\frac{k_*}{k_*} \sum_{y \in S} y + y_* \right) \\
 &= \frac{1}{k_* + 1} (k_* f(x_*) + y_*) \\
 &= \underbrace{\frac{k_*}{k_* + 1}}_{=1-t} f(x_*) + \underbrace{\frac{1}{k_* + 1}}_{=t} y_* \\
 &= (1 - t)f(x_*) + ty_*.
 \end{aligned}$$

Thus the average over T between two values f attains, hence by the intermediate value property, there is an $x_{\dagger} \in I$ such that

$$f(x_{\dagger}) = \frac{1}{k_* + 1} \sum_{y \in T} y.$$

Hence by the principle of mathematical induction, we are done. $\square$

For a counterexample then, it suffices to come up with an example of a function that has the intermediate value property *without* being continuous.

Claim: Let I be an interval of positive length. Then there is a function $f \in F(I; \mathbb{R})$ such that

- f is discontinuous, and
- f has the average value property.

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Proof: Let $\lambda, \rho \in I$, $\lambda \leq \rho$, and consider

$$f : I \rightarrow \mathbb{R}, x \mapsto \begin{cases} \frac{x - \lambda}{\rho - \lambda}, & \text{if } \lambda \leq x \leq \rho \\ 0, & \text{else} \end{cases}$$

The image of this function is $[0, 1]$, and hence by the above claim it satisfies the average value property. However this function is discontinuous at ρ .

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9.1 Follow-up Exercises

- (i) A subset $A \subseteq \mathbb{R}$ is said to be **closed under averages** if for any finite nonempty $S \subseteq A$, $\frac{1}{\#(S)} \sum_{s \in S} s \in A$. Then
- (a) The intersection of $\mathbb{Q}$ with any interval is closed under averages.
 - (b) Let $A \subseteq [0, 1] \cap \mathbb{Q}$. Then A is closed under averages iff $A = [0, 1] \cap \mathbb{Q}$.
- (ii) A real valued function defined on an interval has the intermediate value property iff it has the average value property.

10 PSet 5, Problem 1

Let $I = [0, 1]$.

- (i) If $f_* \in \text{Homeo}(I)$ and $\varepsilon \in \mathbb{R}_{>0}$, then there is a piecewise linear $f \in \text{Homeo}(I)$ such that $d_{C^0}(f, f_*) < \varepsilon$.
- (ii) If $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow \text{Homeo}(I)$ converges uniformly to a function $f_\infty \in F(I; I)$, then $f_\infty \in \text{Homeo}(I)$.
- (iii) If $f_* \in \text{Homeo}(I)$, then there is an $\varepsilon \in \mathbb{R}_{>0}$ such that for any $f \in C^0(I; I)$, if $d_{C^0}(f, f_*) < \varepsilon$, then $f \in \text{Homeo}(I)$.
- (iv) If $f, g \in \text{Homeo}(I)$ are both strictly increasing, then there is a $\Phi \in \text{Homeo}(I)$ such that $\Phi \circ f = g \circ \Phi$.

We study the given statements one by one. First we establish a claim regarding the monotonicity property of homeomorphisms of the unit interval, namely that depending on how a given homeomorphism maps the endpoints, we can determine whether or not it is strictly increasing or strictly decreasing.

Claim: Let $f \in \text{Homeo}(I)$. Then

- (i) $f(\{0, 1\}) = \{0, 1\}$.
- (ii) $f(0) = 0$ iff $f(1) = 1$ iff f is strictly increasing.
- (iii) $f(0) = 1$ iff $f(1) = 0$ iff f is strictly decreasing.

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Proof: (i) For the first part, suppose wlog $f(0) \in]0, 1[$. Then since f is a bijection and hence an injection, either for any $x \in]0, 1]$, $f(x) > f(0)$ or for any $x \in]0, 1]$, $f(x) < f(0)$. In either case, $\text{im}(f)$ is a proper subset of $[0, 1]$, that is, f is not a surjection, a contradiction.

Thus f preserves the **set of endpoints** of I . Since f is bijective, there are exactly two cases:

- $f(0) = 0$ and $f(1) = 1$, corresponding to the second claim, and
- $f(0) = 1$ and $f(1) = 0$, which corresponds to the third claim.

(ii) Suppose $f(0) = 0$. Let $x_1, x_2 \in I$ be such that $x_1 < x_2$ but $f(x_1) > f(x_2)$. But then $f(x_2) \in]0, f(x_1)[$, and since the restriction $f|_{[0, x_1]}$ is continuous, there is an $x_3 \in]0, x_1[$ such that $f(x_3) = f(x_2)$, so that f fails to be an injection, a contradiction. Thus f is strictly increasing. Conversely, if f is strictly increasing, then as $0 < 1$, $f(0) < f(1)$, and since both of these are elements in $\{0, 1\}$, it must be the case that $f(0) = 0$.

(iii) The third part follows from the second part as we can replace f with $f \circ \iota$ for ι the homeomorphism defined by $x \mapsto 1 - x$ ¹⁶.

┘

Next we'll discuss PL approximation of homeomorphisms of the unit interval. This will help us engineer many concrete homeomorphisms that are fairly straightforward to draw and visualize. Here is some heuristics (♥) as to how this can be accomplished: For f a homeomorphism of the unit interval, we consider the diagonal map $f_0 = \text{id}_I : x \mapsto x$ as a first attempt at a PL approximation. Then the function $|f_0 - f|$ itself is a continuous, non-negative function on I , and hence by the Maximum Principle, it attains its maximum at some point $x_1 \in]0, 1[$; note that this maximum is exactly the C^0 distance from f_0 to f . Now replace f_0 with the PL homeomorphism f_1 whose graph consists of the two line segments connecting any consecutive two of the following three points on the graph of f :

¹⁶Precomposing a homeomorphism by ι geometrically corresponds to reflecting the graph of the homeomorphism along the line $x = 1/2$.

$$(0, f(0)), \quad (x_1, f(x_1)), \quad (1, f(1)).$$

Again by the Maximum Principle, there is some point $x_2 \in]0, 1[\setminus \{x_1\}$ where $|f_1 - f|$ is maximized. Again we replace f_1 by the PL homeomorphism f_2 whose graph connects any two consecutive three points (say for simplicity $x_2 > x_1$):

$$(0, f(0)), \quad (x_1, f(x_1)), \quad (x_2, f(x_2)), \quad (1, f(1)).$$

Continuing this way, one ought to be able to approximate f uniformly as accurately as needed. While this method would work, formalization of it is rather cumbersome to write down (without additional groundwork), and we'll instead present a proof that is of a similar flavor, but is less algorithmic¹⁷.

We'll describe the approximating PL homeomorphisms using the notion of a (finite interval) partition of a compact interval. We introduced this notion **later than** the deadline of this problem set, still partitions are not necessary to solve this problem, instead they are (for me) convenient to use, especially because partitions allow one to suppress indices.

Let f be a homeomorphism of I , and $\mathcal{P}$ be a (finite interval) partition of I with each $P \in \mathcal{P}$ of positive length. We define a new function $f_{\mathcal{P}} : I \rightarrow I$ by the condition that for each $P \in \mathcal{P}$, the graph of $f_{\mathcal{P}}$ restricted to P is the line segment from $(\inf(P), f(\inf(P)))$ to $(\sup(P), f(\sup(P)))$. A straightforward calculation gives that the explicit formula for $f_{\mathcal{P}}$ is as follows: for any $P \in \mathcal{P}$ and for any $x \in P$,

$$f_{\mathcal{P}} : x \mapsto m_P x + b_P,$$

¹⁷The algorithm we just described vaguely, while theoretically valid, is not quite practical. For instance with successive steps it is not necessarily true that for any fixed x_0 , the vertical difference along the $x = x_0$ is strictly increasing in general. Efficient approximation of functions using simpler functions is the main focus of **approximation theory**.

where

$$m_P = \frac{f(\sup(P)) - f(\inf(P))}{\sup(P) - \inf(P)},$$

$$b_P = \frac{\sup(P)f(\inf(P)) - \inf(P)f(\sup(P))}{\sup(P) - \inf(P)}.$$

Then by construction f_P is a well defined function from I to I ; further it is piecewise linear and continuous. Next note that by the previous established claim f is either strictly increasing or strictly decreasing, which implies that f_P is either strictly increasing or strictly decreasing, hence by a **theorem proved in class**, f_P is a homeomorphism of I .

Claim: For any $f_* \in \text{Homeo}(I)$ and $\varepsilon \in \mathbb{R}_{>0}$, there is a piecewise linear $f \in \text{Homeo}(I)$ such that

$$d_{C^0}(f, f_*) < \varepsilon.$$

┘

Proof: Let f_* be a homeomorphism of the interval; wlog we may assume that it is strictly increasing. Let further $\varepsilon \in \mathbb{R}_{>0}$. By a **theorem stated (but not proved) in class**, f_* is uniformly continuous. Hence there is a $\delta \in \mathbb{R}_{>0}$ such that for any $x_1, x_2 \in I$,

$$|x_1 - x_2| < \delta \implies |f(x_1) - f(x_2)| < \varepsilon/2.$$

By the Archimedean Principle, there is a positive integer n such that $\delta > 1/n$. Let $\mathcal{P}$ be the finite interval partition of I such that each $P \in \mathcal{P}$ has length exactly $1/n$, and consider the associated PL homeomorphism f_P . We claim that f_P is less than ε away from f relative to the C^0 metric. Indeed

$$\begin{aligned}
d_{C^0}(f_P, f_*) &= \max_{P \in \mathcal{P}} \max_{\inf(P) \leq x \leq \sup(P)} |f_P(x) - f(x)| \\
&= \max_{P \in \mathcal{P}} \max_{\inf(P) \leq x \leq \sup(P)} |m_P x + b_P - f(x)| \\
&\leq \max_{P \in \mathcal{P}} \max_{\inf(P) \leq x \leq \sup(P)} \text{diam}(f(P)) \\
&\leq \max_{P \in \mathcal{P}} \max_{\inf(P) \leq x \leq \sup(P)} \varepsilon/2 \\
&= \varepsilon/2 < \varepsilon.
\end{aligned}$$

┘

The remaining three claims are not quite accurate the way they are stated¹⁸.

Claim: A function from I to itself that is the uniform limit of a sequence of homeomorphisms of I need not be a homeomorphism in general.

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Proof: We'll construct a sequence of PL homeomorphisms that converges uniformly to a PL continuous function that fails to be a homeomorphism. For any $n \in \mathbb{Z}_{\geq 1}$, let $f_n : I \rightarrow \mathbb{R}$ be the PL continuous function whose graph consists of the three line segments connecting any consecutive two of the following four points:

$$(0, 0), \quad \left(\frac{1}{4}, \frac{1}{2} - \frac{1}{n}\right), \quad \left(\frac{3}{4}, \frac{1}{2} + \frac{1}{n}\right), \quad (1, 1).$$

Then for any $n \in \mathbb{Z}_{\geq 3}$, f_n is strictly increasing and onto I , hence is a homeomorphism of I . Similarly, let $f_\infty : I \rightarrow I$ be the PL continuous function whose graph consists of the three line segments connecting any consecutive two of the following four points:

¹⁸Still, all of them can be made true with appropriate, and fairly reasonable, modifications. We list some such modifications as follow-up exercises.

$$(0, 0), \quad \left(\frac{1}{4}, \frac{1}{2}\right), \quad \left(\frac{3}{4}, \frac{1}{2}\right), \quad (1, 1).$$

As f_∞ is not injective, it can not be a homeomorphism. A straightforward calculation shows that for any $n \in \mathbb{Z}_{\geq 3}$, the C^0 distance between f_n and f_∞ is exactly $1/n$, hence $f_\bullet \rightarrow f_\infty$ uniformly.

┘

Claim: For any $\varepsilon \in \mathbb{R}_{>0}$, there is an $f_\varepsilon \in C^0(I; I)$ such that

- f_ε is not a homeomorphism, and
- $d_{C^0}(f_\varepsilon, \text{id}_I) < \varepsilon$.

┘

Proof: For any $n \in \mathbb{Z}_{\geq 3}$, let $f_n : I \rightarrow \mathbb{R}$ be the PL continuous function whose graph consists of the three line segments connecting any consecutive two of the following four points:

$$(0, 0), \quad \left(\frac{1}{2} - \frac{1}{n}, \frac{1}{2}\right), \quad \left(\frac{1}{2} + \frac{1}{n}, \frac{1}{2}\right), \quad (1, 1).$$

None of these functions is injective, hence no f_n is a homeomorphism. By a straightforward calculation, we have that the C^0 distance between f_n and id_I is exactly $1/n$, hence given $\varepsilon \in \mathbb{R}_{>0}$, by the Archimedean Principle we are done.

┘

Claim: Two strictly increasing homeomorphisms of I need not be topologically conjugate in general.

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Proof: Put $I = [0, 1]$ and let $g : [0, 1] \rightarrow [0, 1]$, $x \mapsto x^2$. As for $x_1, x_2 \in I$,

$$|g(x_1) - g(x_2)| = |x_1^2 - x_2^2| = |x_1 + x_2||x_1 - x_2| \leq 2|x_1 - x_2|,$$

g is (**Lipschitz** hence) continuous. g is further strictly increasing with image I , consequently it is a strictly increasing homeomorphism of the unit interval. $f = \text{id}_I$ too is a strictly increasing homeomorphism of I . If there were a topological conjugacy Φ from f to g , we would have

$$\Phi = \Phi \circ f = g \circ \Phi \implies g = \text{id}_I;$$

this is a contradiction since $g(1/2) = 1/4 \neq 1/2$.

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10.1 Follow-up Exercises

- (i) Verify all "straightforward calculations" in the solution presented above.
- (ii) Write explicitly a formula for a PL continuous function that interpolates finitely many points.
- (iii) Let $f \in \text{Homeo}(I)$ and $\iota \in \text{Homeo}(I)$ be defined by $x \mapsto 1 - x$. Then
 - (a) f is strictly increasing iff there is a continuous $\gamma : I \rightarrow \text{Homeo}(I)$ such that $\gamma(0) = \text{id}_I$ and $\gamma(1) = f$.
 - (b) f is strictly decreasing iff there is a continuous $\gamma : I \rightarrow \text{Homeo}(I)$ such that $\gamma(0) = \iota$ and $\gamma(1) = f$.
 - (c) f is strictly increasing iff $f \circ \iota$ is strictly decreasing.
- (iv) Formalize the heuristics (♥).
- (v) For sequences of PL continuous functions, uniform convergence is equivalent to convergence of their **control points**. More precisely; let I be a compact interval, $m \in \mathbb{Z}_{\geq 1}$,

$$x_{1,\bullet}, x_{2,\bullet}, \dots, x_{m,\bullet} : \mathbb{Z}_{\geq 0} \cup \{\infty\} \rightarrow I,$$

$$y_{1,\bullet}, y_{2,\bullet}, \dots, y_{m,\bullet} : \mathbb{Z}_{\geq 0} \cup \{\infty\} \rightarrow \mathbb{R}$$

be such that for any $n \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$,

$$0 \leq x_{1,n} \leq x_{2,n} \leq \dots \leq x_{m,n} \leq 1.$$

Further, for any $n \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, let $f_n : I \rightarrow \mathbb{R}$ be the PL continuous function whose graph consists of the line segments

connecting any consecutive two of the following points:

$$(0, 0), \quad (x_{1,n}, y_{1,n}), \quad (x_{2,n}, y_{2,n}), \quad \dots \quad (x_{m,n}, y_{m,n}), \quad (1, 1).$$

Then $f_\bullet \rightarrow f_\infty$ uniformly iff for any $i \in \{1, 2, \dots, m\}$, $x_{i,\bullet} \rightarrow x_{i,\infty}$ and $y_{i,\bullet} \rightarrow y_{i,\infty}$.

(vi) Let $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow \text{Homeo}(I)$ be a sequence of functions and $f_\infty \in F(I; I)$. Suppose $f_\bullet \rightarrow f_\infty$ uniformly. Then

(a) If the sequence $f_\bullet^{-1}$ of functional inverses is uniformly convergent, then f_∞ is a homeomorphism and f_∞^{-1} is the uniform limit of $f_\bullet^{-1}$.

(b) If f_∞ is a homeomorphism, then the sequence $f_\bullet^{-1}$ of functional inverses converges uniformly to f_∞^{-1} .

(vii) For any $f \in \text{Homeo}(I)$ and any $\varepsilon \in \mathbb{R}_{>0}$, there is an $f_\varepsilon \in C^0(I; I)$ such that

- f_ε is not a homeomorphism, and
- $d_{C^0}(f_\varepsilon, f) < \varepsilon$.

(viii) $\delta : (f, g) \mapsto d_{C^0}(f, g) + d_{C^0}(f^{-1}, g^{-1})$ is a metric on the collection $\text{Bij}(I)$ of all bijections from I to itself, as well as $\text{Homeo}(I)$.

(ix) Composition of functions is continuous on $F_b(I; I)$, that is, if $f_\bullet \rightarrow f_\infty$ and $g_\bullet \rightarrow g_\infty$ are two uniformly convergent sequences of bounded functions, then $g_\bullet \circ f_\bullet \rightarrow g_\infty \circ f_\infty$ uniformly.

(x) Composition of functions is continuous on $\text{Homeo}(I)$ relative to both d_{C^0} and δ .

- (xi) Inversion of functions is discontinuous relative to the C^0 metric but continuous relative to δ .
- (xii) For any $\alpha \in \mathbb{R}_{>0}$, let $e_\alpha : I \rightarrow I, x \mapsto x^\alpha$. Then
- (a) $e_\bullet : \mathbb{R}_{>0} \rightarrow \text{Homeo}(I)$ is a continuous family of strictly increasing homeomorphisms.
 - (b) For any $\alpha, \beta \in \mathbb{R}_{>0}$: $\alpha \neq 1 \neq \beta$ iff e_α and e_β are topologically conjugate.
- (xiii) Let $f, g \in \text{Homeo}(I)$. Then
- (a) If f and g are topologically conjugate, then $f(0) = g(0)$.
 - (b) If $f(0) = g(0)$ and $\text{Fix}(f) \cap]0, 1[= \emptyset = \text{Fix}(g) \cap]0, 1[$, then f and g are topologically conjugate.
- (xiv) Let $f, g \in \text{Homeo}(I)$ be such that $f(0) = g(0)$, and $\Phi \in \text{Homeo}(I)$. Then Φ is a topological conjugacy from f to g iff
- (a) $\Phi(\text{Fix}(f)) = \text{Fix}(g)$ and
 - (b) For any interval $J \subseteq I \setminus \text{Fix}(f)$ and for any $x \in [\inf(J), \sup(J)]$, $\Phi \circ f(x) = g \circ \Phi(x)$.

11 PSet 6, Problem 5

Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow R(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$.

Consider the following statements:

I: $f_\bullet \rightarrow f_\infty$ uniformly.

II: $f_\bullet \rightarrow f_\infty$ pointwise.

III: f_∞ is Riemann integrable.

IV: f_∞ is Riemann integrable and

$$\lim_{n \rightarrow \infty} \int_I f_n(x) dx = \int_I f_\infty(x) dx.$$

V: For each $n \in \mathbb{Z}_{\geq 0}$, $|f_n - f_\infty|$ is Riemann integrable and

$$\lim_{n \rightarrow \infty} \int_I |f_n(x) - f_\infty(x)| dx = 0.$$

Then

- (i) If **I**, then **IV** and **V**.
- (ii) If **II**, then **III**.
- (iii) If **II** and **III**, then **IV**.
- (iv) If **II** and **IV**, then **V**.

All but the first statement are false the way they are stated, although they can be modified to obtain correct versions. In a sense, the types of counterexamples we'll present here are exactly the obstructions for correct statements. We start with consequences of

uniform convergence.

Claim (I $\implies$ III): Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow R(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$. If $f_\bullet \rightarrow f_\infty$ uniformly, then $f_\infty \in R(I; \mathbb{R})$.

┘

Proof: Let $f_\bullet : I \rightarrow \mathbb{R}$ be a sequence of Riemann integrable functions converging uniformly to the function $f_\infty : I \rightarrow \mathbb{R}$. We claim that f_∞ is Riemann integrable. Let us use the aperture characterization of Riemann integrability. As Darboux apertures are defined only for bounded functions; so we first prove that f_∞ is bounded. By uniform convergence, choosing $\varepsilon = 1$, there is an N such that

$$d_{C^0}(f_N, f_\infty) \leq 1.$$

But then for any $x \in I$, by the triangular inequality we have

$$\begin{aligned} |f_\infty(x)| &\leq |f_\infty(x) - f_N(x)| + |f_N(x)| \\ &\leq d_{C^0}(f_N, f_\infty) + \sup(|f_N|) \\ &\leq 1 + \sup(|f_N|). \end{aligned}$$

f_N is Riemann integrable, hence bounded (a **fact we proved in class**), hence $\sup(|f_\infty|) \leq 1 + \sup(|f_N|) < \infty$, so that f_∞ too is bounded.

Next we prove that f_∞ is Riemann integrable. Let $\mathcal{P} \in \text{Par}(I)$, $P \in \mathcal{P}$ and $x_1, x_2 \in P$. Then for n arbitrary we have

$$\begin{aligned} &|f_\infty(x_1) - f_\infty(x_2)| \\ &\leq |f_\infty(x_1) - f_n(x_1)| + |f_n(x_1) - f_n(x_2)| + |f_n(x_2) - f_\infty(x_2)| \\ &\leq 2d_{C^0}(f_n, f_\infty) + |f_n(x_1) - f_n(x_2)| \\ &\leq 2d_{C^0}(f_n, f_\infty) + \text{diam}(f_n(P)). \end{aligned}$$

Thus we have

$$\text{diam}(f_\infty(P)) \leq 2d_{C^0}(f_n, f_\infty) + \text{diam}(f_n(P)).$$

(Note that this bound also implies that f_∞ must be bounded, as there are only finitely many elements P in $\mathcal{P}$.)

Taking the Darboux apertures we obtain, for any $\mathcal{P}$ and any n :

$$\begin{aligned} D(f_\infty; \mathcal{P}) &\leq \sum_{P \in \mathcal{P}} [2d_{C^0}(f_n, f_\infty) + \text{diam}(f_n(P))] \ell(P) \\ &= 2d_{C^0}(f_n, f_\infty) \ell(I) + D(f_n; \mathcal{P}). \end{aligned}$$

Let $\varepsilon \in \mathbb{R}_{>0}$. Then by the uniform convergence hypothesis there is an $N = N(\varepsilon)$ such that

$$2d_{C^0}(f_N, f_\infty) \ell(I) < \varepsilon/2.$$

f_N is Riemann integrable, thus by the aperture characterization of Riemann integrability, there is a partition $\mathcal{P} = \mathcal{P}(\varepsilon, N) \in \text{Par}(I)$ such that

$$D(f_N; \mathcal{P}) < \varepsilon/2.$$

Thus for this particular partition $\mathcal{P}$, we have

$$D(f_\infty; \mathcal{P}) \leq 2d_{C^0}(f_N, f_\infty) \ell(I) + D(f_N; \mathcal{P}) < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

┘

Now that we have guaranteed the Riemann integrability of the uniform limit of a sequence of Riemann integrable functions, we move on to the convergence of the Riemann integrals.

Claim (I $\implies$ IV and V): If $f_\bullet \rightarrow f_\infty$ uniformly on I and $f_\bullet$ is a sequence of Riemann integrable functions, then

$$(i) \int_I |f_\bullet(x) - f_\infty(x)| dx \rightarrow 0.$$

$$(ii) \int_I f_\bullet(x) dx \rightarrow \int_I f_\infty(x) dx.$$

┘

Proof: We first note that by an **exercise stated in class**, for any n , $|f_n - f_\infty|$, is Riemann integrable and further, it suffices to verify the first statement, as for any n ,

$$\left| \int_I f_n(x) dx - \int_I f_\infty(x) dx \right| \leq \int_I |f_n(x) - f_\infty(x)| dx.$$

Note that for any $x \in I$, $|f_n(x) - f_\infty(x)| \leq d_{C^0}(f_n, f_\infty)$, hence by the monotonicity of Riemann integrals, we have

$$0 \leq \int_I |f_n(x) - f_\infty(x)| dx \leq d_{C^0}(f_n, f_\infty) \ell(I).$$

Taking limits as $n \rightarrow \infty$, we are done.

┘

Next we study the remaining statements¹⁹. The main idea is that with pointwise convergence, the chain of implications that works in the case of uniform convergence may fail at every single step (without additional assumptions).

Claim (II $\not\Rightarrow$ III): Let I be a compact interval of positive length. Then there are $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow R(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$ such that

- $f_\bullet \rightarrow f_\infty$ pointwise, but
- $f_\infty \notin R(I; \mathbb{R})$.

¹⁹As you are reading through the counterexamples, drawing sketches of the functions is recommended.

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Proof: Wlog we may assume $I = [0, 1]$. Define $f_\infty : I \rightarrow \mathbb{R}$ by $x \mapsto 1$ if $x \in \mathbb{Q}$ and 0 otherwise. We showed in class that this function is not Riemann integrable. Thus it suffices to construct a sequence of Riemann integrable functions that converges pointwise to f_∞ .

To build such a sequence, we first construct a sequence that visits any rational number in I exactly once. Note that

$$I \cap \mathbb{Q} = \bigcup_{b \in \mathbb{Z}_{\geq 1}} \left\{ 0 = \frac{0}{b}, \frac{1}{b}, \frac{2}{b}, \dots, \frac{b-1}{b}, \frac{b}{b} = 1 \right\}.$$

Thus the sequence $\tilde{q}_\bullet : \mathbb{Z}_{\geq 1} \rightarrow I$ defined by

$$\begin{aligned} \tilde{q}_1 &= 0/1 = 0, \tilde{q}_2 = 1/1 = 1, \\ \tilde{q}_3 &= 0/2 = 0, \tilde{q}_4 = 1/2, \tilde{q}_5 = 2/2 = 1, \\ \tilde{q}_6 &= 0/3 = 0, \tilde{q}_7 = 1/3, \tilde{q}_8 = 2/3, \tilde{q}_9 = 3/3 = 1, \\ \tilde{q}_{10} &= 0/4 = 0, \tilde{q}_{11} = 1/4, \tilde{q}_{12} = 2/4 = 1/2, \tilde{q}_{13} = 3/4, \tilde{q}_{14} = 4/4 = 1, \\ &\dots \end{aligned}$$

has a subsequence $q_\bullet : \mathbb{Z}_{\geq 1} \rightarrow I$ that is a bijection onto $I \cap \mathbb{Q}$. For any $n \in \mathbb{Z}_{\geq 1}$, define $f_n : I \rightarrow \mathbb{R}$ by

$$x \mapsto \begin{cases} 1, & \text{if } x \in \{q_k \mid k \leq n\} \\ 0, & \text{else} \end{cases}.$$

A straightforward calculation gives that for any n , f_n is Riemann integrable with zero integral. Further, for x irrational, $f_n(x) = 0 = f_\infty(x)$ and for x rational, there is a unique $N \in \mathbb{Z}_{\geq 1}$ such that $q_N = x$, hence for any $n \geq N$, $f_n(x) = 1 = f_\infty(x)$, so that $f_\bullet \rightarrow f_\infty$ pointwise.

┘

Claim (II and III $\not\Rightarrow$ IV): Let I be a compact interval of positive length. Then there are $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow R(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$ such that

- $f_\bullet \rightarrow f_\infty$ pointwise,
- $f_\infty \in R(I; \mathbb{R})$, but
- $\lim_{n \rightarrow \infty} \int_I f_n(x) dx \neq \int_I f_\infty(x) dx$.

┘

Proof: Again we may assume $I = [0, 1]$. Define $f_\infty : I \rightarrow \mathbb{R}$, $x \mapsto 0$. Then f_∞ is Riemann integrable with vanishing Riemann integral. We'll construct a sequence of step functions (which are consequently Riemann integrable) that converges pointwise to f_∞ , but the numerical sequence their integrals does not converge to 0. Indeed, for $n \in \mathbb{Z}_{\geq 0}$, define $f_n : I \rightarrow \mathbb{R}$ to be the step function that is constantly 2^n on the subinterval $\left[1 - \frac{1}{2^n}, 1 - \frac{1}{2^{n+1}}\right]$ and zero otherwise. Then by a straightforward calculation for each n , the integral of f_n is 1, and consequently the sequence of integrals is constantly 1, and so does not converge to the integral of f_∞ .

┘

Claim (II and IV $\Rightarrow$ V): Let I be a compact interval of positive length. Then there are $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow R(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$ such that

- $f_\bullet \rightarrow f_\infty$ pointwise,
- $f_\infty \in R(I; \mathbb{R})$,
- $\lim_{n \rightarrow \infty} \int_I f_n(x) dx = \int_I f_\infty(x) dx$, but
- $\lim_{n \rightarrow \infty} \int_I |f_n(x) - f_\infty(x)| dx \neq 0$.

┘

Proof: We'll modify the previous counterexample to ensure convergence of integrals but not the integrals of the absolute values. Wlog assume $I = [0, 1]$. Define $f_\infty : I \rightarrow \mathbb{R}$ by $x \mapsto 0$. For any $n \in \mathbb{Z}_{\geq 0}$, let $f_n : I \rightarrow \mathbb{R}$ be the step function defined so that it is constantly 2^n on the subinterval $\left[1 - \frac{1}{2^n}, 1 - \frac{3/2}{2^{n+1}}\right]$, constantly -2^n on the subinterval $\left[1 - \frac{3/2}{2^{n+1}}, 1 - \frac{1}{2^{n+1}}\right]$, and zero otherwise. Again by a straightforward calculation, for each n , the integral of f_n is zero, hence the sequence of integrals converges to the integral of the limit function f_∞ , however for any n , $|f_n - f_\infty|$ is the n -th step function in the previous example, hence the sequence of integrals of *this* sequence of functions does not converge to zero.

┘

11.1 Follow-up Exercises

- (i) Verify all calculations in the above solutions.
- (ii) Write a closed formula for the n -th term of $\tilde{q}_\bullet$ as well as of $q_\bullet$.
- (iii) (Monotone Convergence Theorem for Riemann Integration) Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow R(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$. If

- $f_\bullet \rightarrow f_\infty$ pointwise,
- $f_\infty \in R(I; \mathbb{R})$, and
- for any $x \in I$, $f_\bullet(x)$ is nondecreasing,

then

- $\lim_{n \rightarrow \infty} \int_I |f_n(x) - f_\infty(x)| dx = 0$, and
- $\lim_{n \rightarrow \infty} \int_I f_n(x) dx = \int_I f_\infty(x) dx$.

- (iv) (Monotone Convergence Theorem for Henstock-Kurzweil Integration) Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow HK(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$. If

- $f_\bullet \rightarrow f_\infty$ pointwise,
- for any $x \in I$, $n \mapsto f_n(x)$ is nondecreasing,
- $n \mapsto \int_I f_n(x) dx$ converges to a finite real number,

then

- $f_\infty \in HK(I; \mathbb{R})$, and
- $\lim_{n \rightarrow \infty} \int_I |f_n(x) - f_\infty(x)| dx = 0$, and
- $\lim_{n \rightarrow \infty} \int_I f_n(x) dx = \int_I f_\infty(x) dx$.

(v) (Bounded Convergence Theorem for Riemann Integration) Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow R(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$. If

- $f_\bullet \rightarrow f_\infty$ pointwise,
- $f_\infty \in R(I; \mathbb{R})$, and
- $f_\bullet$ is a bounded sequence relative to the C^0 metric,

then

- $\lim_{n \rightarrow \infty} \int_I |f_n(x) - f_\infty(x)| dx = 0$, and
- $\lim_{n \rightarrow \infty} \int_I f_n(x) dx = \int_I f_\infty(x) dx$.

(vi) (Bounded Convergence Theorem for Henstock-Kurzweil Integration) Let I be a compact interval, $f_\bullet : \mathbb{Z}_{\geq 0} \rightarrow HK(I; \mathbb{R})$, $f_\infty \in F(I; \mathbb{R})$. If

- $f_\bullet \rightarrow f_\infty$ pointwise, and
- $\inf_n f_n, \sup_n f_n \in HK(I; \mathbb{R})$,

then

- $n \mapsto \int_I f_n(x) dx$ converges to a finite real number,
- $f_\infty \in HK(I; \mathbb{R})$,
- $\lim_{n \rightarrow \infty} \int_I |f_n(x) - f_\infty(x)| dx = 0$, and
- $\lim_{n \rightarrow \infty} \int_I f_n(x) dx = \int_I f_\infty(x) dx$.

12 PSet 7, Problem 3

Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$ with $\lambda < \rho$.

(i) For any $f \in R(I; \mathbb{R})$, there is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) = \frac{1}{\ell(I)} \int_I f(x) dx.$$

(ii) For any $f \in C^0(I; \mathbb{R})$, there is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) = \frac{1}{\ell(I)} \int_I f(x) dx.$$

(iii) For any $f \in C^0(I; \mathbb{R})$ and for any $g \in R(I; \mathbb{R})$, there is an $x_* \in]\lambda, \rho[$ such that

$$\int_I f(x)g(x)dx = f(x_*) \int_I g(x)dx.$$

(iv) For any $f \in C^0(I; \mathbb{R})$ and for any $g \in R(I; \mathbb{R}_{\geq 0})$, there is an $x_* \in]\lambda, \rho[$ such that

$$\int_I f(x)g(x)dx = f(x_*) \int_I g(x)dx.$$

This problem on the one hand connects to [PSet 4, Problem 1](#) above and on the other hand to the mean value theorem(s) in differential calculus. Indeed, the kinds of statements given in this problem are often called mean value theorems for integral calculus. The first and third statements are not quite correct, and indeed we may consider the second and fourth statements as corrected versions of them, respectively. Note further that the constant function $x \mapsto 1$ is Riemann

integrable, and if in the fourth statement we take g to be this function, we recover the second statement. Thus after verifying that the first and third statements are incorrect the way they are stated, it will suffice to verify the fourth statement.

Claim: Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$ with $\lambda < \rho$. Then there is a function $f \in F(I; \mathbb{R})$ such that

- (i) f is a step function (hence is Bourbaki, hence Riemann integrable),
- (ii) The integral (hence average) of f is 0,
- (iii) 0 is not in the image of f .

┘

Proof: Let $\mu = (\lambda + \rho)/2$ be the midpoint of I , and define $f : I \rightarrow \mathbb{R}$ by $x \mapsto -1$ if $x \leq \mu$ and $x \mapsto 1$ otherwise. Then f is a step function, hence it is Bourbaki and Riemann integrable; its integral and average is zero, but it never attains the value zero.

┘

Next we study the third statement. It is not quite correct the way it is given, and we'll break it by constructing two functions f, g such that fg has nonzero integral but g has zero integral. For g we'll take the step function we used before.

Claim: Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$ with $\lambda < \rho$. Then there are functions $f, g \in F(I; \mathbb{R})$ such that

- (i) f is continuous,
- (ii) g is a step function (hence Riemann integrable),
- (iii) The integral of g is zero,

(iv) The integral of fg is not zero²⁰.

┘

Proof: Let $\mu = (\lambda + \rho)/2$ be the midpoint of I , and define $g : I \rightarrow \mathbb{R}$ by $x \mapsto -1$ if $x \leq \mu$ and $x \mapsto 1$ otherwise. Then g is a step function, hence it is Bourbaki and Riemann integrable. Further the integral of g is zero.

Next, define $f : I \rightarrow \mathbb{R}$ by $x \mapsto 0$ if $x \leq \mu$ and $x \mapsto x$ otherwise. Then f is a continuous function, and

$$\int_I f(x)g(x)dx = \int_{[\mu, \rho]} xdx \geq 0.$$

┘

Note that in the proof above, in the last step all we needed was to ensure that the integral of the product is nonzero. For the specific example we had, we could have approximated the integral from below via the integral of a step function, or we could have referred to the following lemma. This lemma is not really necessary, although it is of separate interest. Throughout we'll use various properties of Riemann integrals given as an **exercise in class**.

Lemma: Let $\phi : I \rightarrow \mathbb{R}$ be a Riemann integrable function defined on a closed and bounded interval $I = [\lambda, \rho]$ with positive length. If

- for any $x \in I$, $\phi(x) \geq 0$ and
- there is some point $\mu \in I$ where ϕ is continuous²¹,

²⁰Note that any continuous function is Bourbaki integrable by a problem in a previous problem set, Bourbaki integrability implies Riemann integrability by an exercise given in class, and the product of two Riemann integrable functions is Riemann integrable by a **discussion in class**.

²¹This assumption actually is redundant; indeed, the **Lebesgue-Vitali Theorem** states that a bounded function defined on a closed and bounded interval is Riemann integrable iff it is continuous off a "negligible" set. Consequently any Riemann integrable function must be continuous at least at some point. The reason we state the lemma with this additional redundant assumption is so that we don't refer to this more advanced result (which for the purposes of this problem is not needed).

then $\int_I \phi(x) dx \geq 0$.

┘

Proof: Since ϕ is continuous at μ , for $\varepsilon \in]0, g(\mu)[$, there is a closed subinterval $J \subseteq I$ containing μ of positive length such that and for any $x \in J$, $|\phi(x) - \phi(\mu)| < \varepsilon$. Thus

$$\forall x \in J : 0 \leq \phi(\mu) - \varepsilon \leq \phi(x).$$

The restriction of ϕ to the subinterval J too is Riemann integrable, and by the monotonicity and interval additivity of Riemann integrals we have

$$\begin{aligned} 0 &\leq \ell(J)(\phi(\mu) - \varepsilon) \\ &= \int_J (\phi(\mu) - \varepsilon) dx \\ &\leq \int_J \phi(x) dx \\ &= \int_{[\lambda, \min(J)]} 0 dx + \int_J \phi(x) dx + \int_{[\max(J), \rho]} 0 dx \\ &\leq \int_{[\lambda, \min(J)]} \phi(x) dx + \int_J \phi(x) dx + \int_{[\max(J), \rho]} \phi(x) dx \\ &= \int_I \phi(x) dx. \end{aligned}$$

┘

Finally we study the remaining two statements:

Claim (MVT for Integral Calculus): Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$

For completeness; here is what it means for a set to be negligible in this context: Let $S \subseteq I$ be a subset and consider the **characteristic function** of S defined by $\chi_S : I \rightarrow \mathbb{R}$, $x \mapsto 1$ if $x \in S$ and $x \mapsto 0$ otherwise. Then S is called **negligible** (or **of zero Lebesgue measure**) if χ_S is HK integrable and its HK integral is zero. Just to give an example, we **proved in class** that $\mathbb{Q} \cap [0, 1]$ is a negligible subset of $[0, 1]$.

with $\lambda < \rho$. Then for any $f \in C^0(I; \mathbb{R})$:

(i) For any $g \in R(I; \mathbb{R}_{\geq 0})$, there is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) \int_I g(x) dx = \int_I f(x) g(x) dx. \quad (\diamond)$$

(ii) There is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) = \frac{1}{\ell(I)} \int_I f(x) dx.$$

┘

Proof: As mentioned above, it suffices to show the first item, as choosing $g : x \mapsto 1$ in the first item gives the second item.

For any $x \in I$, we have

$$\min(f) \leq f(x) \leq \max(f).$$

As $g \geq 0$, we may multiply these inequalities by $g(x)$ to obtain

$$\min(f)g(x) \leq f(x)g(x) \leq \max(f)g(x).$$

Integrating, we obtain

$$\begin{aligned} \min(f) \int_I g(x) dx &= \int_I \min(f)g(x) dx \\ &\leq \int_I f(x)g(x) dx \\ &\leq \int_I \max(f)g(x) dx = \max(f) \int_I g(x) dx \end{aligned}$$

Either $\int_I g(x) dx = 0$, or $\int_I g(x) dx > 0$. In the first case, both sides of $(\diamond)$ are zero, and hence we may take as x_* any point in $] \lambda, \rho[$.

Suppose $\int_I g(x)dx \geq 0$; then dividing the terms by this positive number, we obtain

$$\min(f) \leq \frac{\int_I f(x)g(x)dx}{\int_I g(x)dx} \leq \max(f).$$

As f is continuous, by the Intermediate Value Theorem, there is an $x_{\dagger} \in [\lambda, \rho]$ such that

$$f(x_{\dagger}) = \frac{\int_I f(x)g(x)dx}{\int_I g(x)dx}.$$

If $x_{\dagger} \in]\lambda, \rho[$, we are done as we can choose $x_* = x_{\dagger}$. Otherwise, suppose $x_{\dagger} \in \{\lambda, \rho\}$. If f is a constant function, then again we are done (as any point can be taken as x_*). Thus suppose f is not constant. We claim that for some $y_{\dagger}, z_{\dagger} \in]\lambda, \rho[$,

$$f(y_{\dagger}) \leq \frac{\int_I f(x)g(x)dx}{\int_I g(x)dx} \leq f(z_{\dagger}).$$

Suppose otherwise, then for $M = \frac{\int_I f(x)g(x)dx}{\int_I g(x)dx}$, $y \mapsto f(y) - M$ is a continuous function defined on $[\lambda, \rho]$ that is *strictly* positive on $] \lambda, \rho [$. By the Maximum Principle, then, either λ or ρ must be a minimum point for this function, where in fact the value attained is zero. Let $\varepsilon \in \mathbb{R}_{>0}$ be less than $\ell(I)/2$. Integrating against g , we obtain

$$\begin{aligned}
0 &= \int_I f(y)g(y)dy - \int_I f(x)g(x)dx \\
&= \int_I f(y)g(y)dy - \frac{\int_I f(x)g(x)dx}{\int_I g(x)dx} \int_I g(y)dy \\
&= \int_I f(y)g(y)dy - M \int_I g(y)dy \\
&= \int_I (f(y) - M)g(y)dy \\
&= \underbrace{\int_{[\lambda, \lambda+\varepsilon]} (f(y) - M)g(y)dy}_{\geq 0} \\
&\quad + \int_{[\lambda+\varepsilon, \rho-\varepsilon]} (f(y) - M)g(y)dy \\
&\quad + \underbrace{\int_{[\rho-\varepsilon, \rho]} (f(y) - M)g(y)dy}_{\geq 0} \\
&\geq \int_{[\lambda+\varepsilon, \rho-\varepsilon]} (f(y) - M)g(y)dy \\
&\geq \underbrace{\left(\min_{y \in [\lambda+\varepsilon, \rho-\varepsilon]} f(y) - M \right)}_{\geq 0} \underbrace{\int_{[\lambda+\varepsilon, \rho-\varepsilon]} g(y)dy}_{\geq 0}.
\end{aligned}$$

Thus we have that for any $\varepsilon \in \mathbb{R}_{>0}$ with $\varepsilon < \ell(I)/2$,

$$\int_{[\lambda+\varepsilon, \rho-\varepsilon]} g(y)dy = 0.$$

On the other hand, for any such ε ,

$$\begin{aligned}
 \int_I g(x)dx &= \int_I g(x)dx - \underbrace{\int_{[\lambda+\varepsilon, \rho-\varepsilon]} g(y)dy}_{=0} \\
 &= \int_{[\lambda, \lambda+\varepsilon]} g(y)dy + \int_{[\rho-\varepsilon, \rho]} g(y)dy \\
 &\leq 2\varepsilon \sup(g).
 \end{aligned}$$

As g is Riemann integrable, it is bounded and hence its supremum is a finite real number, and hence taking the infimum over ε , we obtain that the integral of g vanishes, a contradiction.

A similar argument gives the existence of a $z_{\dagger}$ as claimed. If $f(y_{\dagger}) = \frac{1}{\ell(I)} \int_I f(x)dx$ or $\frac{1}{\ell(I)} \int_I f(x)dx = f(z_{\dagger})$, we are done. Otherwise again applying the Intermediate Value Theorem to the restriction of f to the closed and bounded interval with endpoints $y_{\dagger}$ and $z_{\dagger}$ guarantees the existence of an x_* as claimed.

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12.1 Follow-up Exercises

- (i) MVT for Integral Calculus can be proved using the Fundamental Theorem(s) of Calculus and (Cauchy's) MVT.
- (ii) Let $I = [\lambda, \rho]$ for some $\lambda, \rho \in \mathbb{R}$ with $\lambda < \rho$. Then for any $f \in C^0(I; \mathbb{R})$ and for any $g \in HK(I; \mathbb{R}_{\geq 0})$,
 - (a) $fg \in HK(I; \mathbb{R})$, and
 - (b) there is an $x_* \in]\lambda, \rho[$ such that

$$f(x_*) \int_I g(x) dx = \int_I f(x)g(x) dx.$$

13 PSet 8, Problem 2

Let I be an open interval, and $x_* \in I$. Then

- (i) For any $\beta \in F(\mathbb{R}; \mathbb{R}_{\geq 0})$, the relation $\|_{x_*, \beta}$ of "being order- β tangent at x_* " is an equivalence relation on $F(I; \mathbb{R})$.
- (ii) If $f, g \in F(I; \mathbb{R})$ are tangent at x_* , then $f(x_*) = g(x_*)$.
- (iii) For any $f \in F(I; \mathbb{R})$, f is differentiable at x_* iff there is exactly one polynomial function of degree at most 1 tangent to f at x_* .
- (iv) For any $f \in F(I; \mathbb{R})$ and for any $n \in \mathbb{Z}_{\geq 0}$, there is at most one polynomial function of degree at most n that is order- β_n tangent to f at x_* , where $\beta_n : h \mapsto h^n$.
- (v) For any $f \in F(I; \mathbb{R})$ and for any $n \in \mathbb{Z}_{\geq 0}$, if $f : I \rightarrow \mathbb{R}$ is n -times differentiable at x_* , then for any n -times differentiable $g \in F(I; \mathbb{R})$ that is order- β_n tangent to f at x_* , where $\beta_n : h \mapsto h^n$, for any $k \in \{0, 1, \dots, n\}$: $f^{(k)}(x_*) = g^{(k)}(x_*)$.

The first statement is true without any modifications:

Claim: "Being order- β tangent at x_* " is an equivalence relation on $F(I; \mathbb{R})$.

┘

Proof: Let us abbreviate "being tangent at x_* " by $\|_{x_*, \beta}$, so that the relation in question formally is

$$\mathcal{R}_{x_*, \beta} = \{(f, g) \in F(I; \mathbb{R}) \times F(I; \mathbb{R}) \mid f \|_{x_*, \beta} g\}.$$

We need to verify three properties; namely reflexivity, symmetry and transitivity.

For reflexivity, let $f \in F(I; \mathbb{R})$. Then as $0 = o(\beta(h))$ ²², $f(x_* + h) = f(x_* + h) + o(\beta(h))$, so that $f \parallel_{x_*, \beta} f$.

For symmetry, let $f, g \in F(I; \mathbb{R})$. Then

$$\begin{aligned} f \parallel_{x_*, \beta} g &\Leftrightarrow f(x_* + h) = g(x_* + h) + o(\beta(h)) \\ &\Leftrightarrow \lim_{h \rightarrow 0} \frac{f(x_* + h) - g(x_* + h)}{\beta(h)} = 0 \\ &\Leftrightarrow \lim_{h \rightarrow 0} \frac{g(x_* + h) - f(x_* + h)}{\beta(h)} = 0 \\ &\Leftrightarrow g(x_* + h) = f(x_* + h) + o(\beta(h)) \Leftrightarrow g \parallel_{x_*, \beta} f. \end{aligned}$$

Finally for transitivity, let $f, g, k \in F(I; \mathbb{R})$ be such that

$$f \parallel_{x_*, \beta} g \parallel_{x_*, \beta} k.$$

We need to show that $f \parallel_{x_*, \beta} k$. This is indeed true since

$$\begin{aligned} f(x_* + h) &= g(x_* + h) + o(\beta(h)) \\ &= k(x_* + h) + o(\beta(h)) + o(\beta(h)) \\ &= k(x_* + h) + o(\beta(h)). \end{aligned}$$

where in the last equation we used the fact that the sum of two functions that decay faster than $\beta(h)$ does as $h \rightarrow 0$, too decays faster than $\beta(h)$ does as $h \rightarrow 0$.

┘

The second statement is not quite correct as given:

Claim: There are functions $f, g : I \rightarrow \mathbb{R}$ tangent at x_* , such that

²²Throughout we abbreviate the more explicit $o_{h \rightarrow 0}$ by o as it is clear from context that the asymptotics are as $h \rightarrow 0$.

$$f(x_*) \neq g(x_*).$$

┘

Proof: Consider for instance $I =] - 1, 1[$, and $f, g : I \rightarrow \mathbb{R}$ defined by

$$f : x \mapsto x, \quad g : x \mapsto \begin{cases} x, & \text{if } x \neq 0 \\ 1, & \text{if } x=0 \end{cases}.$$

Finally put $x_* = 0$. Then for any $h \neq 0$, $f(x_* + h) = g(x_* + h)$, hence **a fortiori**, $f(x_* + h) = g(x_* + h) + o(h)$, so that f and g are tangent at x_* , but these two functions don't take the same value at x_* .

┘

Note that in the counterexample we gave above one of the functions is discontinuous. Indeed, by assuming continuity at x_* , one can fix the second statement:

Claim: For any $f, g \in F(I; \mathbb{R})$, if f and g are both continuous at x_* , and $f \parallel_{x_*} g$ then $f(x_*) = g(x_*)$.

┘

Proof: Let $f, g : I \rightarrow \mathbb{R}$ be two functions continuous at x_* . Note that we can describe the continuity of these two functions by

$$f(x_* + h) = f(x_*) + o(1), \quad g(x_* + h) = g(x_*) + o(1),$$

as $o(1) = o_{h \rightarrow 0}(1)$ represents a function that decays to zero as $h \rightarrow 0$ (albeit possibly not faster than h itself does). Suppose further that f is tangent to g at x_* . We'll use the **epsilon of room principle** to show that $f(x_*) = g(x_*)$, that is, we'll show that $|f(x_*) - g(x_*)|$ is bounded by an expression that is arbitrarily small:

$$\begin{aligned}
|f(x_*) - g(x_*)| &\leq |f(x_*) - f(x_* + h)| \\
&\quad + |f(x_* + h) - g(x_* + h)| \\
&\quad + |g(x_* + h) - g(x_*)| \\
&\leq o(1) + o(h) + o(1) \\
&= o(1).
\end{aligned}$$

┘

For the next statement, we'll again need to include the continuity assumption:

Claim: If $f : I \rightarrow \mathbb{R}$ is a function, then f is differentiable at x_* iff f is continuous at x_* and there is exactly one function $\ell : I \rightarrow \mathbb{R}$ of the form $\ell : x \mapsto Ax + B$ for $A, B \in \mathbb{R}$ such that $f \parallel_{x_*} \ell$.

┘

Proof: ($\Rightarrow$) If f is differentiable at x_* , then it is continuous at x_* (as discussed in class). Further, putting $A = f'(x_*)$ and $B = f(x_*)$ defines an affine function tangent to f at x_* , and by the previous claim this is the only affine function tangent to f at x_* .

($\Leftarrow$) If f is continuous at x_* and $\ell : x \mapsto Ax + B$ is the (an) affine function tangent to f at x_* , then as ℓ is a polynomial it is continuous, and so by a previous claim $f(x_*) = Ax_* + B$. Then

$$\begin{aligned}
f(x_* + h) &= \ell(x_* + h) + o(h) \\
&= (Ax_* + B) + Ah + o(h) \\
&= f(x_*) + Ah + o(h);
\end{aligned}$$

This is equivalent to the statement that f is differentiable at x_* , and

$$f'(x_*) = A.$$

┘

On the other hand, without the continuity assumption, the **sufficiency** breaks down:

Claim: There is a function $f : I \rightarrow \mathbb{R}$ such that there is a unique affine function $\ell : I \rightarrow \mathbb{R}$ tangent to f at x_* , however f is not differentiable at x_* .

┘

Proof: We can build on the previous counterexample we gave. Consider $I =] - 1, 1[$, $x_* = 0$ and $f, g : I \rightarrow \mathbb{R}$ defined by

$$f : x \mapsto x, \quad g : x \mapsto \begin{cases} x, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}.$$

Then f is an affine function and is tangent to g at x_* as discussed in the previous counterexample. Any other affine function ℓ tangent to g at x_* must also be tangent to f at x_* by the first claim, and we'll show in the next claim that $\ell = f$. Consequently there is a unique affine function tangent to g at x_* , however g is not continuous at x_* , let alone differentiability.

┘

The two remaining statements focus on higher order tangencies. We first study the fourth statement:

Claim: (i) For any $n \in \mathbb{Z}_{\geq 0}$, if two polynomial functions are order- β_n tangent at some point, then they are equal as functions.

(ii) For any $f \in F(I; \mathbb{R})$ and for any $n \in \mathbb{Z}_{\geq 0}$, there is at most one polynomial function of degree at most n that is order- β_n tangent to f at x_* , where $\beta_n : h \mapsto h^n$.

┘

Proof: Previously we had shown that order- β tangency is an equivalence relation, hence the first statement immediately implies the second one. To prove the first statement, we'll use induction on the degree n of the polynomials. For $n = 0$, let $p : x \mapsto a_0$ and $q : x \mapsto b_0$ be constant polynomials that are order- β_0 tangent at some point x_* . Then

$$\begin{aligned} 0 &= \lim_{h \rightarrow 0} \frac{|p(x_* + h) - q(x_* + h)|}{|h|^0} \\ &= \lim_{h \rightarrow 0} |a_0 - b_0| \\ &= |a_0 - b_0|. \end{aligned}$$

Next, suppose the statement holds for $n = n_* \in \mathbb{Z}_{\geq 0}$, and let $p : x \mapsto \sum_{k=0}^{n_*+1} a_k x^k$ and $q : x \mapsto \sum_{k=0}^{n_*+1} b_k x^k$ be two polynomials of degree at most $n_* + 1$ that are order- β_{n_*+1} tangent at some point x_* . Then

$$\begin{aligned} &\frac{1}{|x - x_*|^{n_*}} \left| \sum_{k=0}^{n_*} a_k x^k - \sum_{k=0}^{n_*} b_k x^k \right| \\ &\leq \frac{1}{|x - x_*|^{n_*}} \left(|p(x) - q(x)| + |a_{n_*+1} - b_{n_*+1}| |x|^{n_*+1} \right) \\ &= |x - x_*| \frac{|p(x) - q(x)|}{|x - x_*|^{n_*+1}} + |a_{n_*+1} - b_{n_*+1}| \frac{|x|^{n_*+1}}{|x - x_*|^{n_*}}, \end{aligned}$$

and hence by the induction hypothesis, for any $k \in \{0, 1, \dots, n_*\}$, $a_k = b_k$. Thus $(p - q) : x \mapsto (a_{n_*+1} - b_{n_*+1})x^{n_*+1}$, and again order- β_n tangency at a point x_* gives that $a_{n_*+1} = b_{n_*+1}$. By the Principle of Mathematical Induction, we are done.

┘

The fifth statement is almost correct the way it is stated; one possible issue is that for $n = 0$, the hypothesis given is 0-times differentiability; instead of this we'll need continuity^{23 24}.

Claim: For any $f \in C^0(I; \mathbb{R})$ and for any $n \in \mathbb{Z}_{\geq 0}$, if $f : I \rightarrow \mathbb{R}$ is n -times differentiable at x_* , then for any n -times differentiable $g \in C^0(I; \mathbb{R})$ that is order- β_n tangent to f at x_* , where $\beta_n : h \mapsto h^n$, for any $k \in \{0, 1, \dots, n\}$: $f^{(k)}(x_*) = g^{(k)}(x_*)$. $\square$

Proof: First note that by the definition of differentiability, there is an open subinterval $J \subseteq I$ containing x_* such that restricted to J , we have the following:

- $f, f', f'', \dots, f^{(n-1)}$ exist and are continuous on J ,
- $f^{(n-1)}$ is differentiable at x_* ,
- $g, g', g'', \dots, g^{(n-1)}$ exist and are continuous on J ,
- $g^{(n-1)}$ is differentiable at x_* .

We'll argue step by step that the derivatives of f and g match on the nose at x_* . For the zeroth order derivatives (that is, the values of the functions themselves) we have:

²³Presumably an alternative approach is to *define* 0-times differentiability as continuity, and not take it to be equivalent to non-differentiability.

²⁴We also note that the claim follows *immediately* from the previous claim, *provided that one can refer to the Taylor Approximation Theorem*; more specifically the Peano form of the remainder. Since we had not covered Taylor Approximation before this problem set was due, we'll need to argue using more primitive tools.

$$\begin{aligned}
|f(x_*) - g(x_*)| &\leq |f(x_*) - f(x_* + h)| \\
&\quad + |f(x_* + h) - g(x_* + h)| \\
&\quad + |g(x_* + h) - g(x_*)| \\
&\stackrel{(\dagger)}{=} o(1) \\
&\quad + o(1) \\
&\quad + o(1) \\
&= o(1),
\end{aligned}$$

where in the equality with $(\dagger)$, we used the continuity of f and g for the first and third estimates and the order- β_0 tangency for the second estimate. Consequently we have

$$\boxed{f(x_*) = g(x_*)}.$$

For the equality of the first derivatives, we have, for $h \neq 0$:

$$\begin{aligned}
|f'(x_*) - g'(x_*)||h| &\leq |f(x_* + h) - [f(x_*) + f'(x_*)h]| \\
&\quad + |g(x_* + h) - [g(x_*) + g'(x_*)h]| \\
&\quad + |f(x_* + h) - g(x_* + h)| \\
&\quad + \underbrace{|f(x_*) - g(x_*)|}_{=0} \\
&\leq o(|h|) \\
&\quad + o(|h|) \\
&\quad + o(|h|^n) \\
&= o(|h|),
\end{aligned}$$

hence $|f'(x_*) - g'(x_*)| = o(1)$, and consequently again

$$\boxed{f'(x_*) = g'(x_*)}.$$

Next we study derivatives of order at least 2. For brevity, put $\phi = f - g$ and $\psi : x \mapsto (x - x_*)^n$. Then by order- n tangency of f and g at x_* , we have

$$\lim_{x \rightarrow x_*} \frac{\phi(x)}{\psi(x)} = 0.$$

Furthermore, a straightforward calculation shows

$$0 = \psi(x_*) = \psi'(x_*) = \psi''(x_*) = \cdots = \psi^{(n-1)}(x_*).$$

Let $x \in J$ be distinct from x_* . Then by the Cauchy Mean Value Theorem applied to ϕ and ψ on the open interval $J(x_*, x)$ with endpoints x_* and x , we have that

$$\begin{aligned} \exists x_1 = x_1(x) \in J(x_*, x) : \\ \psi'(x_1)(\underbrace{\phi(x) - \phi(x_*)}_{=0}) &= \phi'(x_1)(\underbrace{\psi(x) - \psi(x_*)}_{=0}); \end{aligned}$$

rearranging and simplifying we obtain:

$$\frac{\phi(x)}{\psi(x)} = \frac{\phi'(x_1)}{\psi'(x_1)}.$$

Next, applying the CMVT to ϕ' and ψ' on $J(x_*, x_1)$, we have:

$$\begin{aligned} \exists x_2 = x_2(x_1) \in J(x_*, x_1) : \\ \psi''(x_2)(\underbrace{\phi'(x_1) - \phi'(x_*)}_{=0}) &= \phi''(x_2)(\underbrace{\psi'(x_1) - \psi'(x_*)}_{=0}); \end{aligned}$$

again rearranging we obtain

$$\frac{\phi(x)}{\psi(x)} = \frac{\phi'(x_1)}{\psi'(x_1)} = \frac{\phi''(x_2)}{\psi''(x_2)}$$

Note that we know:

- The limit of these fractions as x approaches x_* is well defined and is zero,
- x_2 is a function of x_1 , which in turn is a function of x , and as x approaches x_* , both x_1 and x_2 too will approach x_* , and consequently
- $\phi'(x_*) = \lim_{x \rightarrow x_*} \phi'(x_2)$, and
- $0 = \psi'(x_*) = \lim_{x \rightarrow x_*} \psi'(x_2)$.

Thus the only way all these can be true is if

$$\phi''(x_2) = o_{x \rightarrow x_*}(|\psi''(x_2)|) = o_{x \rightarrow x_*}(|x_2 - x_*|^{n-2}),$$

which then forces that $\phi''(x_*) = 0$, that is,

$$\boxed{f''(x_*) = g''(x_*)}.$$

We can continue in this way, applying CMVT to higher derivatives of ϕ and ψ on smaller and smaller intervals, so that we have:

$$\begin{aligned}
& \forall x \in J \setminus \{x_*\}, \\
& \exists x_1 = x_1(x) \in J(x_*, x), \\
& \exists x_2 = x_2(x_1) \in J(x_*, x_1), \\
& \dots \\
& \exists x_{n-1} = x_{n-1}(x_{n-2}) \in J(x_*, x_{n-2}) : \\
& \frac{\phi(x)}{\psi(x)} \\
& = \frac{\phi'(x_1)}{\psi'(x_1)} \\
& = \frac{\phi''(x_2)}{\psi''(x_2)} \\
& = \dots \\
& = \frac{\phi^{(n-1)}(x_{n-1})}{\psi^{(n-1)}(x_{n-1})}
\end{aligned}$$

The argument for equality for the second derivatives at x_* works for higher order derivatives with $\leq n$ almost verbatim. The equality of the highest order derivatives, note that the last ratio we obtained is the ratio whose limit is the n -th derivative of ϕ up to some nonzero constant, evaluated at x_{n-1} . Since by hypothesis this derivative exists, again by taking x to x_* , we have

$$0 = \lim_{x \rightarrow x_*} \frac{\phi(x)}{\psi(x)} = \lim_{x \rightarrow x_*} \frac{\phi^{(n-1)}(x_{n-1})}{\psi^{(n-1)}(x_{n-1})} = \phi^{(n)}(x_*)/n!,$$

thus we can conclude that

$$\boxed{f^{(n)}(x_*) = g^{(n)}(x_*)}.$$

└

13.1 Follow-up Exercises

(i) Put for $f, g \in F(I; \mathbb{R})$,

$$f \parallel'_{x_*} g \Leftrightarrow \limsup_{h \rightarrow 0} \frac{f(x_* + h) - g(x_* + h)}{h} < \infty,$$

$$f \parallel''_{x_*} g \Leftrightarrow \limsup_{h \rightarrow 0} \frac{f(x_* + h) - g(x_* + h)}{h} = 0,$$

where for a real valued function ϕ ,

$$\limsup_{h \rightarrow 0} \phi(h) = \inf_{\delta \in \mathbb{R}_{>0}} \sup_{h \in]0, \delta[} \phi(h).$$

Then

- (a) $\parallel'_{x_*}$ and $\parallel''_{x_*}$ too are equivalence relations on $F(I; \mathbb{R})$.
- (b) If $f, g \in F(I; \mathbb{R})$, then $f \parallel''_{x_*} g$ iff $f \parallel'_{x_*} g$ iff $f \parallel_{x_*} g$.
- (ii) Let $\beta_\bullet : \mathbb{Z}_{\geq 0} \rightarrow F(I; \mathbb{R}_{\geq 0})$, $f_\bullet, g_\bullet : \mathbb{Z}_{\geq 0} \cup \{\infty\} \rightarrow F(I; \mathbb{R})$, $x_\bullet : \mathbb{Z}_{\geq 0} \cup \{\infty\} \rightarrow I$ be such that $\beta_\bullet \rightarrow \beta_\infty$ uniformly, $f_\bullet \rightarrow f_\infty$ uniformly, $g_\bullet \rightarrow g_\infty$ uniformly and $x_\bullet \rightarrow x_\infty$. If for any $n \in \mathbb{Z}_{\geq 0}$, $f_n \parallel_{x_n, \beta_n} g_n$, then $f_\infty \parallel_{x_\infty, \beta_\infty} g_\infty$.
- (iii) The relation "being tangent up to order β at a point" makes sense for functions between metric spaces.
- (iv) Prove the last claim proved above *without* using any tools downstream from Rolle Lemma, including the Mean Value Theorem and Cauchy's Mean Value Theorem, as well as Taylor approximation.

14 PSet 8, Problem 5

Let I be a nonempty open interval, and $f \in F(I; \mathbb{R})$. Then

- (i) If f is Newton integrable, then it is continuous.
- (ii) If f is Newton integrable, then it takes any value between any two values it takes (More explicitly, for any subinterval $J \subseteq I$, the image $f(J)$ is an interval).
- (iii) If f is Newton integrable and $f^{-1}(0) = \emptyset$, then either for any $I = f^{-1}(\mathbb{R}_{>0})$ xor $I = f^{-1}(\mathbb{R}_{<0})$.
- (iv) If f takes any value between any two values it takes, then it is Newton integrable.

The first statement given is not correct; indeed if it were, any differentiable function would be continuously differentiable, which is not true. The most straightforward example for this is the **function discussed in class that is Newton integrable (hence locally HK integrable), but not even Riemann integrable**.

Claim: The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$x \mapsto \begin{cases} 2x \cos(1/x^2) + \frac{2}{x} \sin(1/x^2), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}.$$

has the following properties simultaneously:

- f is Newton integrable,
- f is discontinuous at 0.

┘

Proof: It's straightforward that the limit $\lim_{x \rightarrow 0} f(x)$ does not exist, hence f is discontinuous at 0. On the other hand, as we discussed in class, f has an antiderivative, that is, it is Newton integrable.

┘

By adjusting the domain of the example in the above claim, we may construct examples of Newton integrable and discontinuous functions on any nonempty open interval:

Claim: Let I be a nonempty open interval. Then there is a real valued, Newton integrable function on I that is not locally Lebesgue integrable (hence is discontinuous).

┘

Proof: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function described in the proof of the previous claim, and let further $x_* \in I$. Then $x \mapsto f(x - x_*)$ is a Newton integrable function on I that fails to be locally Lebesgue integrable.

┘

Claim: Let I be a nonempty open interval, and $f \in F(I; \mathbb{R})$. Then

- (i) If f is Newton integrable, then it takes any value between any two values it takes (More explicitly, for any subinterval $J \subseteq I$, the image $f(J)$ is an interval).
- (ii) If f is Newton integrable and $f^{-1}(0) = \emptyset$, then either for any $I = f^{-1}(\mathbb{R}_{>0})$ xor $I = f^{-1}(\mathbb{R}_{<0})$.

┘

Proof: Note that the second item follows immediately from the first item; indeed, if f is Newton integrable, then by the first item $f(I)$ is an interval. If in addition f never vanishes, so that $0 \notin f(I)$, then $f(I) \subseteq \mathbb{R}_{>0}$ or $f(I) \subseteq \mathbb{R}_{<0}$. Thus it suffices to prove the first item.

Let $F : I \rightarrow \mathbb{R}$ be an antiderivative of f and let $x_1, x_2 \in I$. Wlog we

may assume $x_1 < x_2$, and $y_1 = f(x_1) < f(x_2) < y_2$. Let $y_3 \in]y_1, y_2[$. Define $\phi : I \rightarrow \mathbb{R}$ by

$$x \mapsto F(x) - y_3 x.$$

Then ϕ is differentiable (and consequently continuous), and $x_* \in I$ is a critical point of ϕ iff $f(x_*) = y_3$. Let $\varepsilon \in \mathbb{R}_{>0}$ be such that $2\varepsilon < x_2 - x_1$. Then applying the Maximum Principle to the restriction of ϕ to $[x_1 + \varepsilon, x_2 - \varepsilon]$, we have that ϕ has a local extremum $x_* \in [x_1 + \varepsilon, x_2 - \varepsilon]$. By a theorem we proved in class, this point is a critical point of ϕ , and consequently we obtained a point $x_* \in]x_1, x_2[$ with $f(x_*) = y_3$. ┘

Claim: The function $g : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$x \mapsto \begin{cases} \frac{1}{x} \cos(\log(|x|)), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}.$$

has the following properties simultaneously:

- g has the intermediate value property, but
 - g is not Newton integrable.
- ┘

Proof: First consider the function $G : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ defined by

$$x \mapsto \sin \circ \log(|x|).$$

Then $G \in C^\infty(\mathbb{R} \setminus \{0\}; \mathbb{R})$ and a straightforward calculation shows that where G is defined, $G' = g$.

We claim that g has the Intermediate Value Property, but it is not Newton integrable. First note that as $G' = g$ on $\mathbb{R} \setminus \{0\}$, for G an infinitely

differentiable function, g is continuous on $\mathbb{R} \setminus \{0\}$, and hence for any interval J that does not contain 0, $g(J)$ is an interval by the Intermediate Value Theorem. Next suppose J is an interval that contains 0. WLOG we may assume that $J = [\lambda, \rho]$ for some $\lambda \leq 0 \leq \rho$. As g is an **odd function**, we may further assume that $\lambda = 0 \leq \rho$ with $g(\rho) > 0$. Fix $y_* \in]0, g(\rho)[$. A straightforward calculation shows that the zero level set of g consists infinitely many points that accumulate to 0 exponentially fast. Let $z_\bullet : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{R}_{>0}$ be a strictly decreasing sequence of zeros of g that converges to 0. Between any two zeros of g , the sign of g is constant. Hence for any n , $g([z_{n+2}, z_n]) \subseteq g([z_{n+4}, z_{n+2}])$ and these subsets constitute a nondecreasing sequence of compact intervals. As $z_\bullet$ decreases to zero, there is a unique n_* such that $\rho \in [z_{n_*+2}, z_{n_*}]$. Consequently

$$y_* \in [0, g(\rho)] \subseteq g([z_{n_*+2}, z_{n_*}]) \subseteq g([z_{n_*+4}, z_{n_*+2}]) \subseteq g([0, \rho]) \subseteq g(J),$$

so that g satisfies the Intermediate Value Property.

Finally we show that g has no antiderivatives. Suppose for a contradiction there is a differentiable function $\Gamma : \mathbb{R} \rightarrow \mathbb{R}$ such that $\Gamma' = g$. Then, as on each interval, a function whose derivative vanishes must be constant, there are two numbers $C_+, C_- \in \mathbb{R}$ such that for any $x \in \mathbb{R}_{>0}$

$$\Gamma(x) = G(x) + C_+, \quad \Gamma(-x) = G(-x) + C_-.$$

As Γ is differentiable everywhere, it is also continuous everywhere, hence in particular it is continuous at 0. However a straightforward calculation shows that the (one or two-sided) limit of $G(x)$ as x approaches 0 does not even exist, a contradiction.

┘

14.1 Follow-up Exercises

- (i) On any nonempty open interval, there are real valued Newton integrable functions that are discontinuous everywhere.
- (ii) Consider the following function from problem 3 of **problem set 4**:

$$\mathbb{R} \rightarrow \mathbb{R}, \quad x \mapsto \begin{cases} \sin(1/x), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}.$$

This function is Newton integrable (but discontinuous).

15 PSet 9, Problem 2

Let $p, q \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$, $f \in C^p(\mathbb{R}; \mathbb{R})$, and $g \in C^q(\mathbb{R}; \mathbb{R})$. Then

- (i) $g \circ f \in C^s(\mathbb{R}; \mathbb{R})$, where $s = \min\{p, q\}$.
- (ii) f is a diffeomorphism iff it is a C^p diffeomorphism.
- (iii) For any $k \in \{0, 1, \dots, s\}$, where $s = \min\{p, q\}$, the k -th derivative of $g \circ f$ can be written as the sum of scalar multiples of products of lower order derivatives of g pre-composed with f and lower order derivatives of f .

All three statements are correct the way they are stated. We start with an auxiliary lemma regarding the regularity of products of two functions; this will be handy as ultimately in what follows we'll use the chain rule, which involves products of functions.

Claim (Auxiliary Lemma): $\forall p \in \mathbb{Z}_{\geq 1} \cup \{\infty\}, \forall A, B \in C^p : AB \in C^p$.

┘

Proof: Note that it suffices to prove the statement for finite p , as

$$C^\infty = \bigcap_{n \in \mathbb{Z}_{\geq 0}} C^n.$$

For $p = 1$, if $A, B \in C^1$, then $A', B' \in C^0$, and since the product of two continuous functions is continuous, and the sum of two continuous functions is also continuous, we have $(AB)' = A'B + AB' \in C^0$. Since the derivative of AB is continuous, this by definition means that $AB \in C^1$. Next say for $p = p_* \in \mathbb{Z}_{\geq 1}$, the product two C^{p_*} functions is C^{p_*} , and suppose $A, B \in C^{p_*+1}$. Then $(AB)' = A'B + AB'$. By induction hypothesis, $A'B, AB' \in C^{p_*}$, hence so is their sum, and consequently $AB \in C^{p_*+1}$. Thus by the Principle of Mathematical induction, the

product of two C^p functions is C^p for any $p \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$. ┘

Note that the above claim can be generalized to products of finitely many functions by induction on the number of factors.

Claim: Let $p, q \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$, $f \in C^p(\mathbb{R}; \mathbb{R})$, and $g \in C^q(\mathbb{R}; \mathbb{R})$. Then

$$g \circ f \in C^s(\mathbb{R}; \mathbb{R}),$$

where $s = \min\{p, q\}$. ┘

Proof: Note that since

$$C^\infty \subseteq \dots \subseteq C^{n+1} \subseteq C^n \subseteq C^1 \subseteq C^0 \subseteq F_{b, \text{loc}} \subseteq F,$$

wlog we may assume $f, g \in C^p$. We'll show that $g \circ f \in C^p$ also. Further, as before it suffices to prove the statement for finite p .

For $p = 1$, if $f, g \in C^1$, then $(g \circ f)' = g' \circ f'$. Composition of two continuous functions is continuous, so $g' \circ f$ is continuous, and $f' \in C^0$ by hypothesis. Thus $(g \circ f)'$ is the product of two continuous functions, so that it is continuous, thus $g \circ f \in C^1$. Next suppose for $p = p_* \in \mathbb{Z}_{\geq 1}$, the composition of two C^{p_*} functions is C^{p_*} . Let $f, g \in C^{p_*+1}$. Then $g', f \in C^{p_*}$, hence $g' \circ f \in C^{p_*}$ by induction hypothesis, and $(g \circ f)' = g' \circ f'$ is the product of two C^{p_*} functions, so that it is C^{p_*} by the auxiliary lemma. Thus $g \circ f \in C^{p_*+1}$. By the Principle of Mathematical Induction, we are done. ┘

Claim: Let $p \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$, $f \in C^p(\mathbb{R}; \mathbb{R})$. Then f is a diffeomorphism iff it is a C^p diffeomorphism. ┘

Proof: Recall that by definition a C^p diffeomorphism is a diffeomorphism that is C^p whose inverse is also C^p . Thus it suffices to assume

- $f \in C^p$,
- f^{-1} exists as a function,
- f^{-1} is differentiable,

and show that $f^{-1} \in C^p$. We know by the inverse function theorem, that the derivative of f is positive everywhere or negative everywhere; wlog the former is the case. Then for $\varphi : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$, $x \mapsto 1/x$, again by the inverse function theorem, we have that

$$(f^{-1})' = \varphi \circ f'.$$

Note that for any $n \in \mathbb{Z}_{\geq 1}$, $\varphi^{(n)} : x \mapsto (-1)^n n! x^{-(n+1)}$ ²⁵, consequently $\varphi \in C^\infty(\mathbb{R}_{>0}; \mathbb{R}_{>0})$, and so $(f^{-1})'$ is the composition of a C^∞ function and a C^{p-1} function, so that by the previous claim, $(f^{-1})' \in C^{p-1}$, hence $f^{-1} \in C^p$.

┘

For the last statement, we'll present a soft argument²⁶. Throughout very little estimates or limits will be involved; instead we'll need to manage the indices and apply various differentiation rules **algebraically**. More specifically, we'll keep track of how complicated higher order derivatives in terms of the number of functions involved. Let $\mathfrak{F} \subseteq F(\mathbb{R}; \mathbb{R})$ (which, we may assume to be a collection of C^∞ functions) and for any $S, P, C, D \in \mathbb{Z}_{\geq 0}$, denote by $\mathfrak{F}(S, P, C, D)$ the collection of all functions $\mathbb{R} \rightarrow \mathbb{R}$ that can be written of the sum of at most S functions, each of which is the product of at most P functions, each of which is the composition of at most C functions, each of which is the some derivative of order at most D of some function in $\mathfrak{F}$.

²⁵Say by induction.

²⁶This is as opposed to a hard argument, where one would derive an exact formula for higher derivatives of compositions. See the Kranks-Parks book, §1.3 (available on Canvas).

Thus for $\mathfrak{F} = \{f, g\}$, we have for instance (we'll use $\cdot$ to denote function product for the sake of sanity and readability):

- $f, g \in \mathfrak{F}(1, 1, 1, 0)$,
- $f', g' \in \mathfrak{F}(1, 1, 1, 1)$,
- $(g \circ f)' = g' \circ f \cdot f' \in \mathfrak{F}(1, 2, 2, 1)$,
- $(g \circ f)'' = (g' \circ f \cdot f')' = (g' \circ f)' \cdot f' + g' \circ f \cdot f'' = g'' \circ f \cdot f' \cdot f' + g' \circ f \cdot f'' \in \mathfrak{F}(2, 3, 2, 2)$ and so on.

Note also that if either one of S, P , or C is 0, then $\mathfrak{F}(S, P, C, D)$ is automatically the emptyset.

Claim: Let $\mathfrak{F} \subseteq C^\infty(\mathbb{R}; \mathbb{R})$, $S, P, C, D \in \mathbb{Z}_{\geq 0}$. Then

$$\forall A \in \mathfrak{F}(S, P, C, D) : A' \in \mathfrak{F}(SP, P + C - 1, C, D + 1).$$

┘

Proof: Wlog, we may assume $S, P, C \geq 1$. Let $A \in \mathfrak{F}(S, P, C, D)$. We may further assume that the first three parameters are all sharp²⁷. Then A is of the form

$$A = \sum_{s=1}^S A_s,$$

where

$$\forall s : A_s = \prod_{p=1}^P A_{sp} \in \mathfrak{F}(1, P, C, D),$$

²⁷In this context this means that A is the sum of *exactly* S functions, each of which is the product of *exactly* P functions, each of which is the composition of *exactly* C functions that are derivatives of functions in $\mathfrak{F}$.

where

$$\forall \mathbf{s}, \mathbf{p} : A_{\mathbf{sp}} = \bigcirc_{c=1}^C A_{\mathbf{spc}} \in \mathfrak{F}(1, 1, C, D),$$

where

$$\forall \mathbf{s}, \mathbf{p}, c : A_{\mathbf{spc}} = f^{(d)} \in \mathfrak{F}(1, 1, 1, D),$$

where $0 \leq d \leq D$ and $f \in \mathfrak{F}$.

First, $A'_{\mathbf{spc}} = f^{(d+1)}$ and $d + 1 \leq D + 1$, which justifies the increment in the bound on number of derivatives involved. Next, by chain rule iterated finitely many times, we have

$$\begin{aligned} A'_{\mathbf{sp}} &= \left(\bigcirc_{c=1}^C A_{\mathbf{spc}} \right)' \\ &= \left(A_{\mathbf{sp}1} \circ A_{\mathbf{sp}2} \circ \cdots \circ A_{\mathbf{sp}C} \right)' \\ &= A'_{\mathbf{sp}1} \circ A_{\mathbf{sp}2} \circ \cdots \circ A_{\mathbf{sp}C} \\ &\quad \cdot A'_{\mathbf{sp}2} \circ A_{\mathbf{sp}3} \circ \cdots \circ A_{\mathbf{sp}C} \\ &\quad \cdot \cdots \\ &\quad \cdot A'_{\mathbf{sp}C} \\ &\in \mathfrak{F}(1, C, C, D + 1). \end{aligned}$$

Next, by product rule iterated finitely many times, we have

$$\begin{aligned}
A'_S &= \left(\prod_{p=1}^P A_{sp} \right)' \\
&= (A_{s1} A_{s2} \cdots A_{sP})' \\
&= A'_{s1} A_{s2} \cdots A_{sP} \\
&\quad + A_{s1} A'_{s2} \cdots A_{sP} \\
&\quad + \cdots \\
&\quad + A_{s1} A_{s2} \cdots A'_{sP} \\
&\in \mathfrak{F}(P, P + C - 1, C, D + 1),
\end{aligned}$$

here we obtained the bound on the number of products based on the fact that each A'_{sp} itself has at most C factors. Finally since we have S many terms as above, we obtain

$$A' \in \mathfrak{F}(SP, P + C - 1, C, D + 1).$$

┘

Claim: Let $p, q \in \mathbb{Z}_{\geq 1} \cup \{\infty\}$, $f \in C^p(\mathbb{R}; \mathbb{R})$, and $g \in C^q(\mathbb{R}; \mathbb{R})$. Then for any $k \in \{0, 1, \dots, s\} \setminus \{\infty\}$, where $s = \min\{p, q\}$, the k -th derivative of $g \circ f$ can be written as

- the sum of at most $k!$ functions, each of which is
- the product of at most $k + 1$ functions, each of which is
- some derivative of f or g of order at most k .

┘

Proof: Put $\mathfrak{F} = \{f, g\}$. Then $g \circ f \in \mathfrak{F}(1, 1, 2, 0)$. We claim that for any $k \leq s$, $(g \circ f)^{(k)} \in \mathfrak{F}(k!, k + 1, 2, k)$. By chain rule we have $(g \circ$

$f)' = g' \circ f \cdot f' \in \mathfrak{F}(1, 2, 2, 1)$. Suppose the statement is correct for $k_* \in \{0, 1, \dots, s - 1\}$, so that $(g \circ f)^{(k_*)} \in \mathfrak{F}(k_*!, k_* + 1, 2, k_*)$. Then $(g \circ f)^{(k_*+1)} = \left((g \circ f)^{(k_*)}\right)' \in \mathfrak{F}((k_* + 1)!, k_* + 2, 2, k_* + 1)$ by the previous claim; thus we are done by the Principle of Mathematical Induction.

┘

15.1 Follow-up Exercises

- (i) Let $r = (p, \theta), s = (q, \psi) \in (\mathbb{Z}_{\geq 0} \cup \{\infty\}) \times]0, 1]$, $f, g \in F(\mathbb{R}; \mathbb{R})$. f is called C^r if it is C^p and its p -th derivative $f^{(p)}$ satisfies a local **θ -Hölder estimate**. If $f \in C^r$ and $g \in C^s$, then $g \circ f \in C^t$, where $t = (m, \omega) \in (\mathbb{Z}_{\geq 0} \cup \{\infty\}) \times]0, 1]$ is defined by

$$m = \min\{p, q\}, \quad \omega = \begin{cases} \theta, & \text{if } m = p \\ \psi, & \text{if } m = q \end{cases}.$$

- (ii) Let $\varphi : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$, $x \mapsto 1/x$. Then for any $n \in \mathbb{Z}_{\geq 0}$, and any $x \in \mathbb{R}_{>0}$, $\varphi^{(n)}(x) = (-1)^n n! x^{-(n+1)}$.
- (iii) $x \mapsto x^3$ is a C^∞ homeomorphism that is not a diffeomorphism.

16 PSet 10, Problem 4

Define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $x \mapsto \begin{cases} \exp(-1/x^2) & , \text{ if } x \neq 0 \\ 0 & , \text{ if } x = 0 \end{cases}$. Then

- (i) For any $n \in \mathbb{Z}_{\geq 1}$, $f(x) = o_{x \rightarrow 0}(x^n)$.
- (ii) f is differentiable at $x = 0$ and $f'(0) = 0$.
- (iii) For any $n \in \mathbb{Z}_{\geq 0}$, there is a polynomial function P_n of degree $3n$ such that the n -th derivative of f satisfies the formula

$$\forall x \in \mathbb{R} \setminus \{0\} : f^{(n)}(x) = P_n\left(\frac{1}{x}\right) f(x).$$

- (iv) For any $n \in \mathbb{Z}_{\geq 0}$, $f^{(n)}(0)$ exists and is zero.
- (v) For any $n \in \mathbb{Z}_{\geq 0}$, $T_n(f; 0)$ is the function that is constantly zero.
- (vi) $f \in C^\infty(\mathbb{R}; \mathbb{R})$.

Definition and Properties of the Exponential Function Put $\epsilon : \mathbb{R} \rightarrow \mathbb{R}_{>0}$, $x \mapsto \exp(x)$, so that

$$f : x \mapsto \begin{cases} \epsilon(-1/x^2) & , \text{ if } x \neq 0 \\ 0 & , \text{ if } x = 0 \end{cases}.$$

There are multiple ways to define the exponential function ϵ , and as such, we may decide to choose any one of the definitions of the exponential function²⁸. We'll use the following definition as it is the most

²⁸See the follow-up exercise regarding different characterizations of the exponential function.

thematic to the theory we developed so far: the function $\mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ defined by $u \mapsto 1/u$ is C^∞ , hence by the second fundamental theorem of calculus, $\mathbb{R}_{>0} \rightarrow \mathbb{R}$, $x \mapsto \int_1^x du/u$ is C^∞ , and it is strictly increasing²⁹. Consequently by the (global) inverse function theorem, it is a C^∞ diffeomorphism; ϵ is *by definition, we take it*, the C^∞ diffeomorphism that is the functional inverse of *this* function. Consequently, for any $x \in \mathbb{R}_{>0}$, we have

$$\epsilon \left(\int_1^x \frac{du}{u} \right) = x.$$

By the chain rule and the second fundamental theorem of calculus, we have, for $x \in \mathbb{R}_{>0}$ arbitrary,

$$\begin{aligned} 1 &= \frac{dx}{dx} \\ &= \frac{d}{dx} \left[\epsilon \left(\int_1^x \frac{du}{u} \right) \right] \\ &= \epsilon' \left(\int_1^x \frac{du}{u} \right) \frac{d}{dx} \left(\int_1^x \frac{du}{u} \right) \\ &= \epsilon' \left(\int_1^x \frac{du}{u} \right) \frac{1}{x} \\ &= \frac{\epsilon' \left(\int_1^x \frac{du}{u} \right)}{\epsilon \left(\int_1^x \frac{du}{u} \right)}, \end{aligned}$$

so that, by the change of variables $y = \int_1^x \frac{du}{u}$,

$$\forall y \in \mathbb{R} : \epsilon'(y) = \epsilon(y) \quad \text{and} \quad \epsilon(0) = 1.$$

Next, applying the product rule we obtain, for any $x \in \mathbb{R}$,

²⁹Of course, this is nothing but the natural logarithm function, although we *won't need to be aware of this fact* for what is to come.

$$\begin{aligned}
 \frac{d}{dx} [\epsilon(x)\epsilon(-x)] &= \epsilon'(x)\epsilon(-x) + \epsilon(x)\frac{d}{dx} [\epsilon(-x)] \\
 &= \epsilon(x)\epsilon(-x) + \epsilon(x)\epsilon'(-x)(-1) \\
 &= 0,
 \end{aligned}$$

so that the function $x \mapsto \epsilon(x)\epsilon(-x)$ is constant on $\mathbb{R}$. As for $x = 0$,

$$[\epsilon(x)\epsilon(-x)]|_{x=0} = \epsilon(0)^2 = 1.,$$

we have that

$$\boxed{\forall x \in \mathbb{R} : \epsilon(-x) = \frac{1}{\epsilon(x)}}. \quad (\dagger)$$

Next, as by the first fundamental theorem of calculus, for any $x \in \mathbb{R}_{\geq 0}$,

$$\epsilon(x) = 1 + \int_0^x \epsilon'(u) du = 1 + \int_0^x \underbrace{\epsilon(u)}_{>0} du \geq 1 = T_0(\epsilon; 0)(x),$$

hence consequently

$$\epsilon(x) = 1 + \int_0^x \epsilon'(u) du = 1 + \int_0^x \underbrace{\epsilon(u)}_{>1} du \geq 1 + x = T_1(\epsilon; 0)(x),$$

and by induction, for any $x \in \mathbb{R}_{\geq 0}$ and any $n \in \mathbb{Z}_{\geq 0}$,

$$\epsilon(x) \geq T_n(\epsilon; 0)(x) \geq \frac{x^n}{n!}.$$

Consequently for any $n \in \mathbb{Z}_{\geq 0}$,

$$\frac{x^n}{e(x)} = \frac{x^n}{x^{n+1}} \frac{x^{n+1}}{e(x)} = \frac{1}{x} (n+1)! \xrightarrow{x \rightarrow \infty} 0,$$

so that

$$\boxed{\forall n \in \mathbb{Z}_{\geq 0} : x^n = o_{x \rightarrow \infty}(e(x))}. \quad (\heartsuit)$$

With all these properties of the exponential function, we can now study the statements given; they are all correct the way they are stated.

Study of Statements

Claim: Let f be the function given above. Then

- (i) For any $n \in \mathbb{Z}_{\geq 0}$, $f(x) = o_{x \rightarrow 0}(x^n)$.
- (ii) f is differentiable at $x = 0$ and $f'(0) = 0$.
- (iii) For any $n \in \mathbb{Z}_{\geq 0}$, there is a polynomial function P_n of degree $3n$ such that the n -th derivative of f satisfies the formula

$$\forall x \in \mathbb{R} \setminus \{0\} : f^{(n)}(x) = P_n\left(\frac{1}{x}\right) f(x).$$

- (iv) For any $n \in \mathbb{Z}_{\geq 0}$, $f^{(n)}(0)$ exists and is zero.
- (v) For any $n \in \mathbb{Z}_{\geq 0}$, $T_n(f; 0)$ is the function that is constantly zero.
- (vi) $f \in C^\infty(\mathbb{R}; \mathbb{R})$.

┘

Proof: Let us first note implications between the different items in the claim:

- Item (i) for $n = 0$ is equivalent to the statement that f is continuous at 0.
- Item (i) for $n = 1$ is equivalent to the statement that f is differentiable at 0 and $f'(0) = 0$, verifying item (ii).
- f is the composition of two infinitely differentiable functions on $\mathbb{R} \setminus \{0\}$, hence is C^∞ there. Thus it would suffice to employ induction on n to verify item (iii); and by taking the limit as x approaches 0, we would establish that f is differentiable at 0 of any order, $f^{(n)}(0) = 0$, and so f is C^∞ , verifying items (iv) and (vi). Consequently any Taylor polynomial of f at 0 is the zero polynomial, verifying item (v).

Consequently it suffices to show items (i) and (iii). We start with item (i). Let $x \neq 0$ and $n \in \mathbb{Z}_{\geq 0}$,

$$\left| \frac{f(x)}{x^n} \right| = \left| \frac{e(-x^{-2})}{x^n} \right| \stackrel{(*)}{=} \left| t^n e(-t^2) \right| \stackrel{(\dagger)}{=} \left| \frac{t^n}{e(t^2)} \right| \xrightarrow{x \rightarrow 0} 0,$$

where in equation $(*)$, we used the substitution $t = 1/x$, and for the limit we use the property $(\heartsuit)$ above.

Next, to establish property (iii), for $n = 1$, we may simply apply the chain rule to obtain:

$$\frac{d}{dx} [f(x)] = \frac{d}{dx} [e(-x^{-2})] = e'(-x^{-2}) 2x^{-3} = f(x) [2s^3] \Big|_{s=1/x},$$

so that we have $P_1(s) = 2s^3$. Next, suppose the statement holds for $n = n_* \in \mathbb{Z}_{\geq 0}$. Then applying the product and chain rules, we obtain:

$$\begin{aligned}
f^{(n_*+1)}(x) &= f^{(n_*)'}(x) \\
&= \frac{d}{dx} [P_{n_*}(1/x)f(x)] \\
&= P'_{n_*}(1/x)(-x^{-2})f(x) + P_{n_*}(1/x)f'(x) \\
&= P'_{n_*}(1/x)(-x^{-2})f(x) + P_{n_*}(1/x)2x^{-3}f(x) \\
&= \left[P'_{n_*}(1/x)(-x^{-2}) + P_{n_*}(1/x)2x^{-3} \right] f(x).
\end{aligned}$$

Note that the first factor is a polynomial; its first summand is a polynomial (in $1/x$) of degree exactly $3n_* + 1$, and the second summand is a polynomial of degree exactly $3n_* + 3 = 3(n_* + 1)$; hence by the principle of mathematical induction, we are done.

┘

As a final note, we comment on the significance of this function. First, this function is infinitely differentiable, hence it has Taylor approximations of any order³⁰. In general, we *would* want the Taylor polynomials to converge to the original function, but in this case, no Taylor polynomial (at 0) can converge to the original function, except exactly at 0. Thus the formal functional series

$$T_\infty(f; 0)(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = \sum_{n=0}^{\infty} 0 = 0,$$

it being constantly zero, *is* convergent, but it fails to converge to $f(x)$, except exactly when $x = 0$!

³⁰Similar functions, called **bump functions** are used to adjust other functions in a smooth manner localized to a certain neighborhood.

16.1 Follow-up Exercises

(i) There is a unique function $e : \mathbb{R} \rightarrow \mathbb{R}_{>0}$ that solves any one of the following **functional equations**:

I: For any $x, y \in \mathbb{R} : e(x + y) = e(x)e(y)$ and e is continuous at some point (alternatively differentiable at some point, or locally integrable etc.).

II: e is differentiable, $e(0) = 1$, and for any $x \in \mathbb{R}$, $e'(x) = e(x)$.

III: For any $x \in \mathbb{R}$, $e(x) = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^{1/n}$.

IV: For any $x \in \mathbb{R}$, $e(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$.

V: For any $x \in \mathbb{R}_{>0}$, $e\left(\int_1^x \frac{du}{u}\right) = x$.

VI: e is locally HK integrable (alternatively locally Riemann integrable etc.) and for any $x \in \mathbb{R}$, $e(x) = 1 + \text{HK} \int_0^x e(u)du$.

17 PSet 11, Problem 2

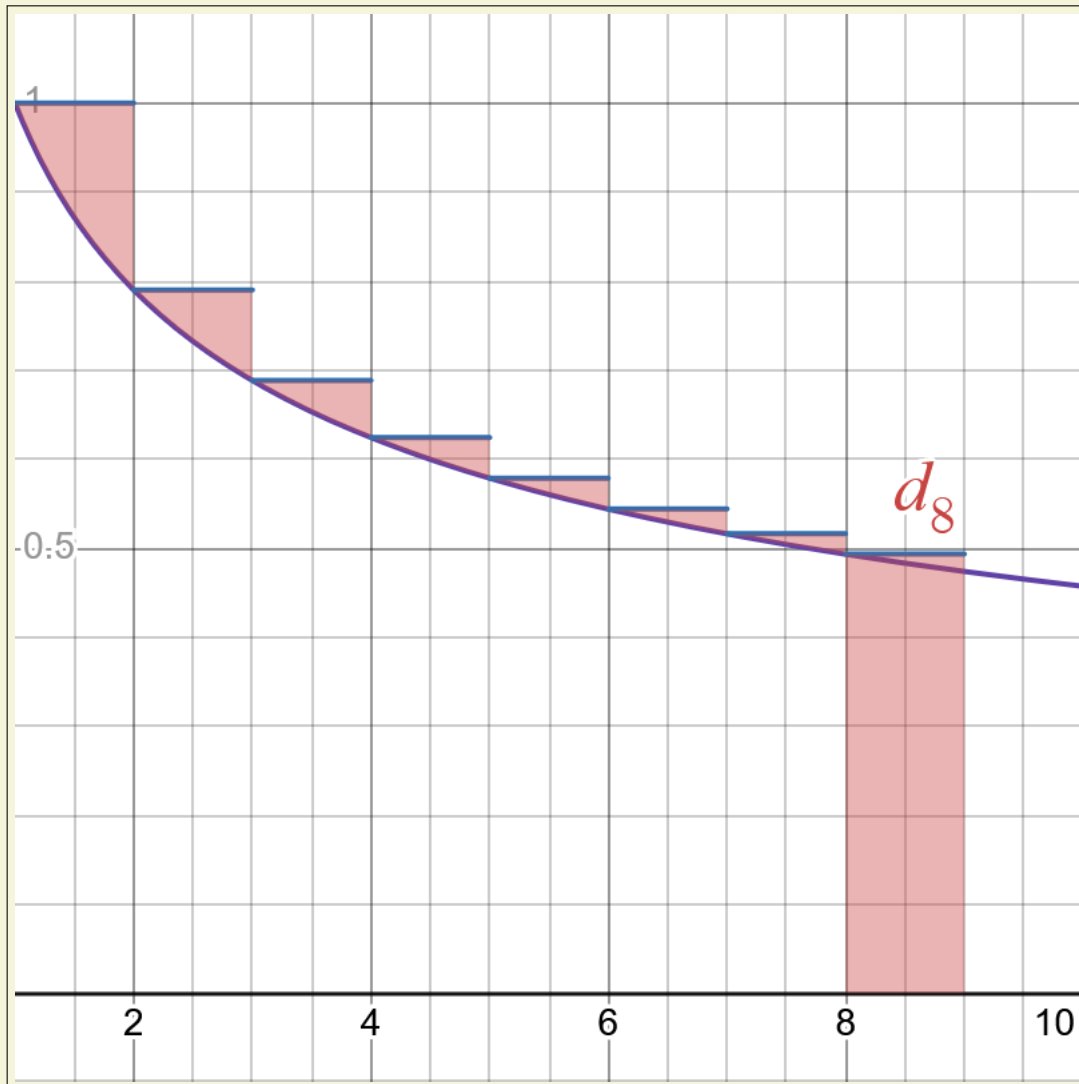
Let $f : [1, \infty[\rightarrow \mathbb{R}_{\geq 0}$ be a nonincreasing function such that $\lim_{x \rightarrow \infty} f(x) = 0$. Define for any $n \in \mathbb{Z}_{\geq 1}$,

$$d_n = \sum_{k=1}^n f(k) - \int_1^n f(x) dx.$$

Then

- (i) For any $n \in \mathbb{Z}_{\geq 1}$, $0 \leq f(n) \leq d_n \leq f(1)$.
- (ii) $d_\bullet$ is nonincreasing.
- (iii) $d_\infty = \lim_{n \rightarrow \infty} d_n$ exists.
- (iv) The formal sum $\sum_{n=1}^{\infty} f(n)$ converges iff the improper integral $\int_1^{\infty} f(x) dx$ converges.
- (v) For any $n \in \mathbb{Z}_{\geq 1}$, $0 \leq d_n - d_\infty \leq f(n)$.
- (vi) $\sum_{k=1}^n f(k) = \int_1^n f(x) dx + d_\infty + O_{n \rightarrow \infty}(f(n))$.

Let us first note that as f is nonincreasing, it is monotone, and by problem 2 of [problem set 5](#), is locally Bourbaki (hence also Riemann, Lebesgue and Henstock-Kurzweil) integrable. Thus the sequence $d_\bullet$ is well defined. Furthermore, for each n , we may consider the finite sum part of d_n as a certain Riemann sum of $f|_{[1, n+1]}$; namely, partition $[1, n+1]$ into intervals of unit length, and sample the left endpoints of each interval in the partition; here is a sketch summarizing this interpretation:



The interactive [Desmos](https://www.desmos.com/calculator/1jihbidmkb) graph the above image is generated from is available at the following link:

<https://www.desmos.com/calculator/1jihbidmkb>.

Besides the intuitive understanding of d_n , this interpretation also hints at the fact that Saks Lemma type bounds will come in handy in the study of this sequence. More specifically, it will be useful to split the integral into a sum of integrals over subintervals.

All given statements are true as given³¹.

³¹Although note that in an earlier version of this problem set, the statements had some unintended typos.

Claim: Let $f : [1, \infty[\rightarrow \mathbb{R}_{\geq 0}$ be a nonincreasing function such that $\lim_{x \rightarrow \infty} f(x) = 0$. Define for any $n \in \mathbb{Z}_{\geq 1}$,

$$d_n = \sum_{k=1}^n f(k) - \int_1^n f(x) dx.$$

Then

- (i) For any $n \in \mathbb{Z}_{\geq 1}$, $0 \leq f(n) \leq d_n \leq f(1)$.
- (ii) d_n is nonincreasing.
- (iii) $d_\infty = \lim_{n \rightarrow \infty} d_n$ exists.
- (iv) The formal sum $\sum_{n=1}^{\infty} f(n)$ converges iff the improper integral $\int_1^{\infty} f(x) dx$ converges.
- (v) For any $n \in \mathbb{Z}_{\geq 1}$, $0 \leq d_n - d_\infty \leq f(n)$.
- (vi) $\sum_{k=1}^n f(k) = \int_1^n f(x) dx + d_\infty + O_{n \rightarrow \infty}(f(n))$.

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Proof: (ii) First note that for any $n \in \mathbb{Z}_{\geq 1}$,

$$\begin{aligned}
d_{n+1} &= \sum_{k=1}^{n+1} f(k) - \int_1^{n+1} f(x) dx \\
&= \sum_{k=1}^n f(k) + f(n+1) - \int_1^n f(x) dx - \int_n^{n+1} f(x) dx \\
&= \sum_{k=1}^n f(k) - \int_1^n f(x) dx - \int_n^{n+1} \underbrace{(f(x) - f(n+1))}_{\geq 0} dx \quad (*) \\
&\leq \sum_{k=1}^n f(k) - \int_1^n f(x) dx \\
&= d_n,
\end{aligned}$$

where in $(*)$ we used the hypothesis that f is nonincreasing. This establishes that d_n is nonincreasing also.

(i) Next, it's immediate that $d_1 = f(1)$, and for any $n \in \mathbb{Z}_{\geq 1}$,

$$\begin{aligned}
d_{n+1} &= \sum_{k=1}^{n+1} f(k) - \int_1^{n+1} f(x) dx \\
&= \sum_{k=1}^{n+1} f(k) - \sum_{k=1}^n \int_k^{k+1} f(x) dx \\
&= f(n+1) + \sum_{k=1}^n \left[f(k) - \int_k^{k+1} f(x) dx \right] \\
&= f(n+1) + \sum_{k=1}^n \int_k^{k+1} \underbrace{(f(k) - f(x))}_{\geq 0} dx \quad (*) \\
&\geq f(n+1),
\end{aligned}$$

where in $(*)$ we again used the fact that f is nonincreasing. Thus

by the Principle of Mathematical Induction, we have:

$$\boxed{\forall n \in \mathbb{Z}_{\geq 0} : f(n) \leq d_n.}$$

Combining this together with the facts that f is a nonnegative function and $d_\bullet$ is nonincreasing, we obtain the chain of inequalities in (i).

(iii) This follows immediately from the first two items by way of the Monotone Convergence Theorem; indeed we also have that

$$\boxed{d_\infty = \lim_{n \rightarrow \infty} d_n \in [0, f(1)]}.$$

(iv) ($\Leftarrow$) Suppose first that the improper integral in question converges, so that by definition the following limit exists:

$$\int_1^\infty f(x)dx = \lim_{\rho \rightarrow \infty} \int_1^\rho f(x)dx \in \mathbb{R}_{\geq 0}.$$

If this limit exists, then by farming we see that we may replace the variable ρ by any sequence that diverges to infinity. Thus for any $n \in \mathbb{Z}_{\geq 1}$,

$$\sum_{k=1}^n f(k) = d_n + \int_1^n f(x)dx \xrightarrow{n \rightarrow \infty} d_\infty + \int_1^\infty f(x)dx.$$

Thus the infinite sum in question converges and

$$\boxed{\sum_{n=1}^\infty f(k) = d_\infty + \int_1^\infty f(x)dx}.$$

($\Rightarrow$) Conversely, suppose the infinite sum converges. For any $\rho \in \mathbb{R}_{\geq 1}$, there is a unique positive integer n_ρ such that $n_\rho \leq \rho < n_\rho + 1$; namely n_ρ is the **floor** of ρ . Note that $\lim_{\rho \rightarrow \infty} n_\rho = \infty$. Then as f is a

nonnegative function,

$$\begin{aligned}
 \sum_{k=1}^{n_\rho} f(k) - d_{n_\rho} &= \int_1^{n_\rho} f(x) dx \\
 &\leq \int_1^{\rho} f(x) dx \\
 &\leq \int_1^{n_{\rho+1}} f(x) dx = \sum_{k=1}^{n_{\rho+1}} f(k) - d_{n_{\rho+1}}
 \end{aligned}$$

Taking the limit as $\rho \rightarrow \infty$, as the sum of two convergent sequences is convergent and converges to the sum of the respective limits, we have by the squeeze theorem that the improper integral converges, and

$$\boxed{\int_1^\infty f(x) dx = \sum_{n=1}^\infty f(n) - d_\infty.}$$

(v) Let $\ell \in \mathbb{Z}_{\geq 1}$. Then

$$\begin{aligned}
 0 \leq d_\ell - d_{\ell+1} &= \left[\sum_{k=1}^{\ell} f(k) - \int_1^{\ell} f(x) dx \right] - \left[\sum_{k=1}^{\ell+1} f(k) - \int_1^{\ell+1} f(x) dx \right] \\
 &= \left[\sum_{k=1}^{\ell} f(k) - \sum_{k=1}^{\ell+1} f(k) \right] + \left[\int_1^{\ell+1} f(x) dx - \int_1^{\ell} f(x) dx \right] \\
 &= -f(\ell+1) + \int_{\ell}^{\ell+1} \underbrace{f(x)}_{\leq f(\ell)} dx \quad (*) \\
 &\leq f(\ell) - f(\ell+1),
 \end{aligned}$$

where again in (*) we used the nonincreasing nature of f , and

that these are all nonnegative terms follows from **(ii)**. Then for any $n, p \in \mathbb{Z}_{\geq 1}$, we have

$$\begin{aligned} 0 \leq d_n - d_{n+p+1} &= \sum_{k=n}^{n+p} (d_k - d_{k+1}) \\ &\leq \sum_{k=n}^{n+p} (f(k) - f(k+1)) = f(n) - f(n+p+1). \end{aligned}$$

Taking the limit $p \rightarrow \infty$, the hypothesis that f vanishes at infinity³², gives that

$$\boxed{\forall n \in \mathbb{Z}_{\geq 1} : 0 \leq d_n - d_{\infty} \leq f(n)}.$$

(vi) This follows immediately from the previous item by the definition of big-Oh. ┘

As a final remark, as this problem relates the convergence or divergence of a certain formal series to that of an improper integral, it is often referred to as the **Integral Test**.

³²Note that this is the first time we used the hypothesis that f vanishes at infinity in this proof.